

NutriScan

Enhancing Consumer Nutrition Literacy Through Technology

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1 Introduction

Having good health is important; it gives people the ability to enjoy and participate in daily activities, achieve their potential and engage in society. It motivates people to contribute more actively to their communities and family on social levels, including work, relationships, and providing support. Likewise, it is also economically beneficial for the society. Therefore, good health is fundamental for both societies and for people individually. Evidently, many factors are included in the assessment of good health, but one of the more important ones is dietary lifestyle (Marshall 2024). It is crucial to have a diet based on a balanced amount of nutrition, which benefits both physical and mental well-being; such as immunity, longevity, disease prevention, and mental clarity. Therefore, a balanced diet is important to function properly as a human being, and according to World Health Organization (WHO) an unbalanced diet can lead to serious risks such as underweight, stunting, malnutrition, obesity, and non-communicable diseases (World Health Organization 2024). The consequences of not eating varied foods can furthermore lead to cardiovascular diseases, strokes, diabetes, or cancer, to which WHO mentions the fact that in 2022 there were registered 2.5 billion overweight adults, 890 million were obese, and 390 million were underweight. In addition to this, WHO mentions that especially women, infants, children, adolescents, and people living in poverty are at risk of different forms of malnutrition (World Health Organization 2024).

Furthermore, WHO describes a healthy diet as following “...fruit, vegetables, legumes (e.g. lentils and beans), nuts and whole grains (e.g. unprocessed maize, millet, oats, wheat and brown rice) [...] Less than 10 % of total energy intake from free sugars [...] Less than 30 % of total energy intake from fats” (World Health Organization 2018, page 2). However, human diets have evolved, leading to a significant increase in processed and ultra-processed foods. What is worrying about these categories of foods are the fact that they often contain high amount of calories, sugars, sodium, and unhealthy fats, and additives, therefore making it hard to get the proper intake of essential nutrients. This shift towards ultra-processed food, affects both public health and sustainability. The NOVA classification system was developed to distinguish categories of processed foods into groups of four, going from unprocessed, such as fresh fruits and vegetables, to ultra-processed foods such as soft drinks, chips, and pre-made foods. The Brazilian epidemiologist Carlos Augusto Monteiro argues that excessive consumption of ultra-processed foods can lead to obesity and non-communicable diseases. In continuation of this, Monteiro points out that an issue also rises with the fact that large food corporations mainly promote ultra-processed foods and therefore influence public policy and behaviour (Monteiro et al. 2018).

Because of the evolution within diets, nutrition labels on packaged foods have become widespread and mandatory. The labels were added to products as a response to dietary changes and as public health intervention to improve consumer awareness of healthy eating. Nutrition labels are meant to give accessible information and encourage buyers to engage in what they consume (Campos, Doxey, and Hammond 2011).

But do the consumers actually read these labels, and if so, do they understand the information and know how to use it? We are curious about this matter, and whether the consumers actively use these labels and have literacy and numeracy skills to understand and process the information. This reflection also emerges from our own experience and motivation, as we find it recognisable from our everyday life that it can be difficult to gain an overview of the exact contents of a product, that seems camouflaged by professional terminology and odd numbers to calculate. In addition to this, we wonder how consumers experience the formulation of the information and whether they feel like the actual content of products is represented clear enough. Furthermore, we wish to uncover how a web solution can facilitate nutrition values and encourage a better understanding of nutrition declarations.

As the reflection above suggests, this wonder and motivation on whether consumers use nutrition declarations and know how to use the information, leads to a deeper research. This introduction does not only explore worrying outcomes of modern dietary behaviour, but also why nutrition labels and health interventions are necessary, as well as their potential as intervention on human health. Furthermore, the introduction lays the groundwork for the problem analysis and our ambition to create a web solution to facilitate understanding on nutrition declarations.

2 Background

In this chapter, there will be an outline of the components of nutrition and its functional impact on health. Furthermore, there will be presented existing knowledge from the field, as well as how our approach aims to differentiate from current practices.

2.1 Nutrition

Nutrition as a concept is not necessarily bound to the human nature, though it is a major component of a human's physical and mental wellbeing. WHO ascribes the strength of our immune systems and overall health to a well-balanced nutritional intake (World Health Organization, n.d.). Nutrition as a term is usually divided between micro and macronutrients, both which are essential for the human body. The macronutrients are those most people are familiar with, such as carbohydrates, proteins and fats, while micronutrients consists of minerals and vitamins (Medicine (U.S.) Panel on Macronutrients. 2005, p. 21-24).

A balanced nutritional intake

The concept of what is established as a healthy diet, is continuously evolving. This aims to incorporate new insights into understanding the roles of the various foods, essential nutrients, and other food components that play into health and disease prevention. A large and expanding body of research supports that the intake of specific types of nutrients, food groups, and dietary patterns can significantly contribute to improving health outcomes and prevent malnutrition and non-communicable diseases such as coronary heart disease, strokes, cardiovascular disease, cancer, and diabetes (Cena and Calder 2020). Dieticians define a healthy, balanced diet as one where macronutrients (i.e., carbohydrates, proteins, and fats) are consumed in appropriate portions to meet the body's needs for growth, remaining healthy and disease prevention. Macronutrients supply the necessary energy required for cellular processes that is needed to support the daily physiological functions, while micronutrients (i.e., vitamins and minerals) are required in smaller quantities to facilitate normal growth, development, metabolism, and overall physiological functioning (Cena and Calder 2020). A healthy diet should not only meet the energy and hydration needs, but also provide an adequate supply of micronutrients, protecting against vitamin or mineral deficiencies, and other nutritional deficiencies, as well as building up the immune system. Through a balanced intake and a nutritious diet, both the energetic and physiological needs will be fulfilled for the body to maintain optimal health (Max Healthcare 2022).

What is the need and functionality of nutrition labels?

The need and functionality of nutrition labels are essential to guarantee that everyone gets the necessary information about nutrition (Fødevarestyrelsen, n.d.). Furthermore it serves to guide the consumers in choices regarding health, and ensuring transparency in food products (KIM, M. NAYGA JR., and CAPPS JR. 2001). That being said, nutrition labels have for most continents and countries become a requirement. For instance, companies selling products in Denmark are commanded to follow the legislation set by Denmark and the European Union's veterinary and food administration (European Commission, n.d.). By 2016, EU made specific requirements to which information is obligated on packages sold in EU-countries. These requirements are, among other things that; information must be accurate, easy to read and understand, including ingredients, date marking, net quality, nutrition labels (European Union 2024).

Nutrition labels are essential to help people make smarter and healthier choices about what they consume. These labels provide important information about the food's ingredients, calories, sugars and fats, helping consumers quickly understand if a food is good for their diet (Feteira-Santos et al. 2020). They also make it easier for people to avoid health risks, like obesity and diabetes, by showing exactly what's in the product. Some products also have s, with symbols or phrases like "low fat" or "high fibre," to make it easier for customers to notice if the product is healthy (Talati et al. 2019). According to WHO, these labels can reduce health risks such as obesity and diabetes, making it easier for people to understand the calories, sugars, fats, and other nutrients that could affect their health. For people who are tracking their weight, focusing on calorie information can help them make healthier choices. For people with allergies, nutrition labels are especially important because they highlight allergens like nuts, dairy, or gluten (Fernandes da Costa Souza and Costa Lima 2016). The European Food Safety Authority argues that transparent labels showing allergens, like nuts or dairy, can help people avoid foods that might trigger health issues (European Food Safety Authority 2024). The ingredient list, which shows everything in the food by weight, also helps people spot any additives or artificial ingredients. These labels show the amounts of nutrients, like carbs, fats, proteins, vitamins and minerals, per serving. This helps people understand if the food fits their dietary needs (Fernandes da Costa Souza and Costa Lima 2016).

Nutrition labels, also play a big role in keeping food companies honest. By law, companies must provide accurate and clear information about what is in their products. This prevents companies from misleading consumers with false claims and therefore helps build trust. It is important that companies provide accurate and reliable information about the contents of their products, as they need to protect the consumer against misleading labelling and marketing (Fødevarestyrelsen, n.d.). Nutrition labels are therefore essential in making food choices easier, healthier, safer, and giving people transparent and accurate information about what is in their food.

2.2 Which problems are already known within the field?

Consumers' use of nutrition labels and their understanding of them is not an uncommon research area. This is likely based on the fact, that there has been a need for health interventions in recent decades, due to changes within human diets. The studies distinguish between nutrition facts tables (NFT) and front-of-package labels (FOPL). NFTs are known by typically being placed on the back of a product and simply through a table that informs consumers of the number of calories, protein, energy, among other things. FOPLs are typically in the front of a product and with only a few words, numbers, and information, explaining the amount of fat, calories, salt. These often use 'traffic light' colours to make it more intuitive where a certain amount is considered low (green), medium (yellow) or high (red) (Hammond et al. 2023).

However, it is difficult to be fully specific about nutrition labels, as studies argue that their visual layout and extent varies according to countries and continents. A study by professor in public health, David Hammond, showed that nutrition information varies across these countries Mexico, Chile, Australia, United Kingdom, Canada, and the United States from guidelines, warning labels, voluntary systems (such as the traffic light label), to FOPL and NFT. Despite the difference, 50-75 % respondents from the surveyed countries, reported seeing nutrition labels often, but interestingly only one in three found them easy to understand (Hammond et al. 2023). Likewise another study, by professor in philosophy Khalid Shamim, examined consumers in India, which showed the fact, that among those who read nutrition labels, only 37.2 % fully understood the information, while a whopping 62.8 % only partially understood the information (Shamim, Khan, and Ahmad 2022).

Furthermore, the study argues that the lack of knowledge or difficult terms (35 %), lack of time (28.3 %), and lack of interest (19.2 %) are the main struggles for the respondents. However the study showed that it was mainly females who always read the labels compared to males, and females mostly looked for fat and sugar content, where males mostly focused on energy and carbohydrates (Shamim, Khan, and Ahmad 2022).

This study was not the only one to recognise a distinction between the genders. Another study by professor in behavioural decision Alexander Persoskie, who researched specifically American consumers, showed that males are more likely to disregard nutrition content and consume sugar-sweetened soda frequently. Furthermore, this study found that people with lower income and shorter education had lower comprehension of nutrition labels. The result showed that participants with longer education scored best on the assignment of understanding nutrition information. Among those with a college degree only 54 % answered correctly to all questions, and among participants with less than a high school diploma 35.4 % were unable to answer any question correctly. Finally, the research paper emphasises how crucial the role of health literacy is, as respondents with higher health literacy manage to understand nutrition labels better (Persoskie, Hennessy, and Nelson 2017).

The study also emphasises, that better education on nutrition is needed and a simplification of NFTs are necessary as part of public health, and encourage informed food choices among consumers (Persoskie, Hennessy, and Nelson 2017). A simplification of the labels will make them more accessible to consumers of all literacy levels and background, which is also recommended in the first mentioned article, where the study showed that mandatory FOPL such as traffic light labels showed more awareness, effectiveness, and engagement among consumers. Likewise, this study emphasises that transparent and intuitive labels will contribute to health outcomes, especially for children and adolescents (Hammond et al. 2023).

2.2.1 Problem Statement

Based on the previous findings and research, it is made noticeable that a large number of consumers struggle to fully understand and make use of nutritional labels, due to: complex terminology, limited time, and varying levels of health literacy. These challenges, paired with the growing need for personalised dietary needs, lead us to the following problem statement:

How can a web solution assist consumers in understanding and navigating within the scientific language of nutritional labels, while offering personal preferences and restrictions that align with dietary needs?

3 Theory and Methodology

The following chapter will explain our selection of theories and methods for this project. In this, we aim to explain how we methodically gathered data to get deeper insights into the problem area and consumer behaviour. In addition we seek to explain how we processed the data and determined user and system requirements.

3.1 Quantitative Method

In the following section, we will dive into the subject of quantitative method and why it is useful for our project and how we used it. In this project, we chose to create a questionnaire to get a deeper insight in who our target demographic are, and what they may think of the problem area. The reason why quantitative method and a questionnaire is a great match for our project is because we can reach a larger population. The responses we acquired through our questionnaire, is an important part of our project, due to the fact that it helps in providing knowledge to cover blind spots concerning the project, that we may have missed. One of the positive things about the quantitative method, is that we can quickly gather many responses from a lot of different people, which can be difficult with the use of qualitative method ([National University 2024](#)). Another positive thing about using quantitative method is, that with a lot of participants, they can be used as a sample, and thereby represent a general opinion of our chosen target demographic. When making a questionnaire a lot of the results from the responses comes out as numeric data, which then easily can be measured, by different statistical measures ([National University 2024](#)).

Some of the downsides of using quantitative method is the lack of depth that the results may have, because of difficulties within defining causation, compared to qualitative method. When focusing on a rather large group of people, like we have done with the questionnaire, we only somewhat get a brief insight into the participants' thought processes and behaviours. This is both from open-ended questions using their own words as well as pre-defined options such as multiple-choice. They do not get the chance to express their thoughts on the subject on a deeper level, since the format is not built for it, thus it may be harder to do over a questionnaire rather than a personal interview. Another thing to be attentive to, is that enough people answer the questionnaire. Otherwise it may be difficult to make a general observation of a target demographic, if there isn't enough participants ([National University 2024](#)).

In our questionnaire we focused on the age group 15-30 years old. We created different age groups, so the respondents would not simply blend together. There are four different age groups in the questionnaire, and they go from 15-18, 19-22, 23-26 and 27-30. We had 63 people answering our questionnaire. In the following diagram it is possible to see how the respondents are distributed between the different age groups, which can be seen in figure 3.1. The questionnaire can be accessed in Appendix 2 for questions and structure, and Appendix 3 for answers.

How old are you?

15-18 years [**blue**]

19-22 years [**red**]

23-26 years [**orange**]

27-30 years [**green**]

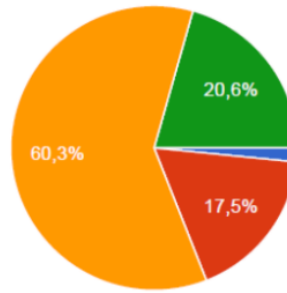


Figure 3.1: Questionnaire

As seen in the figure above it is clear that most of the people who answered the questionnaire belongs to the age group 23-26 years old. 38 people who answered the questionnaire belongs to this group. Beyond this, 13 people belong to the age group, 27-30 years old, while 11 people are in the group 19-22 and only a single respondent is in the group 15-18 years old. Most of the responses were from men, as 43 people were men and 16 of the respondents were women. This could possibly be because, the questionnaire was shared through the Department of Computer Science, therefore it does not necessarily show a correlation between gender and interest in nutrition. To get an idea of our respondents' behaviour of consumption, and what is of importance to them when shopping for groceries, the following are some of the questions they had to answer:

- "What considerations do you make when shopping for groceries?"
- "Describe a "good product" in three or more words"
- "Is there anything that limits your shopping?"
- "Is there anything that limits your shopping?"
- "Is there anything that limits your shopping?"
- "How do you plan your shopping trips (before and during)?"
- "What do you consider as "healthy"?"
- "What comes to mind when you hear the word "nutrition"?"

These were questions in which the participants had to write their answers themselves and where they did not have the opportunity to check boxes with predetermined answers.

3.2 Qualitative Method

Additionally, we have utilised qualitative method in the project, in the form of interviews. We interviewed five people who were in the range of our target demographic, and interviewed them with questions related to our questionnaire. The interviews can be found in Appendix 5. The reason we did this was to get a deeper understanding on who our target demographic is, and as mentioned in the section above, when answering a questionnaire, it can be difficult to fully express ones thoughts, and with an interview this can be rectified. In extension of our questionnaire, our interviews has provided us with a lot of different data, concerning our target demographic, which helps in pinpointing the focus for our project, and what different needs our target demographic express an interest in, when creating our web solution.

Some of the positive things about using qualitative method as a means of gathering data, is the fact that it is easier to interpret the data and interviews than it is with questionnaires. Another positive aspect of the qualitative method is that, if respondents answers surprisingly to a question, we get the chance to ask follow-up questions, through which it is possible to elaborate further. This is something we do not have the option of doing, when working with a questionnaire. One of the downsides of using qualitative method, is that the answers we get from the interviews are only the opinions and experiences of five people. Another aspect, which is important to consider, when working with qualitative method, is the importance of remaining neutral when conducting the interviews, so the participants does not get influenced by our questions. In extension of this, is it also important to not just pick the answers which favours our project, but to consider every aspect of the answers we get ([National University 2024](#)).

The questions we asked our participants when interviewing them was the following:

- Do you look at the nutritional content/ingredient list? Why? What do you pay attention to and why not anything else?
- What could make you want to look at/read them?
- How can the nutritional content be presented more clearly/made more attractive for you to read?
- Are you interested in being informed more about any of the other content?
- Why?/Why not?"

By asking these questions, is it easier to understand the participants dietary and shopping habits. The answers we got from our participants varied, though we discovered similarities which helped us narrowing our target demographic and creating our requirements. Some of the answers from the interviews were as follows:

- *"I think I would be interested in learning a bit more about things I don't know about, like additives and what they actually mean"* (appx. 5, line 327-330).
- *"Maybe if it were presented in a simpler way or with colours, it would actually be much easier to understand. If I could quickly see whether something is healthy or not, I would save time on my shopping trips"* (appx. 5, line 433-436).
- *"...if you tell yourself that you eat healthy, it would actually be nice to know what healthy means"* (appx. 5, line 96-98).

The answers that we got from our participants are very important for our project as they have helped us to create and design our web solution, based on the needs and wants of the participants in regards to their shopping habits.

3.3 Computational Thinking

In this section we seek to explore the components within Computational Thinking (CT), along with their functionality within this project. CT is comprised of four components: *decomposition*, *pattern recognition*, *abstraction* and *algorithm design*. These will each serve a different role, in order to break down complex problems, to better highlight issues and experiences, to which we can tailor specific solutions (Wing 2006). Going forward, we will now explore the functionality of these elements, of which CT is comprised, along with how they have granted us the ability to achieve a new and deeper understanding of the issue at hand.

Decomposition

Decomposition is the first step within CT, where we seek to break down a complex problem, or concept, into more manageable parts, from which each individual part is more easily worked upon (Csizmadia et al. 2015). Within this project, our main focus is to explore possible web solutions to assist the readability of nutrition labels. This would be a broad subject to engage upon, hence the need to narrow down the problem through decomposition. First step of this process is to categorise the problem into smaller parts and figure out different actors and objects and how they engage in activities. This categorisation can be seen in figure 3.2.

Actors	Objects	Activities
Consumers Producers Legislators (DK and EU) Grocery store Governmental health advice	Web Product (unprocessed, processed, and ultra-processed foods) Nutrition Facts Labels Ingredient list Applications	Shopping Reading labels Understanding information on labels Comparing prices Comparing products

Figure 3.2: Decomposition into core elements

These are a few examples of the many subparts of which the problem at hand consists of. Furthermore, we have made an effort to condense these parts into a visualised flowchart, in which we seek to descent the ladder of relevant problems, by drawing lines from our problem statement, to the two main actors identified within this project (see figure 3.3)

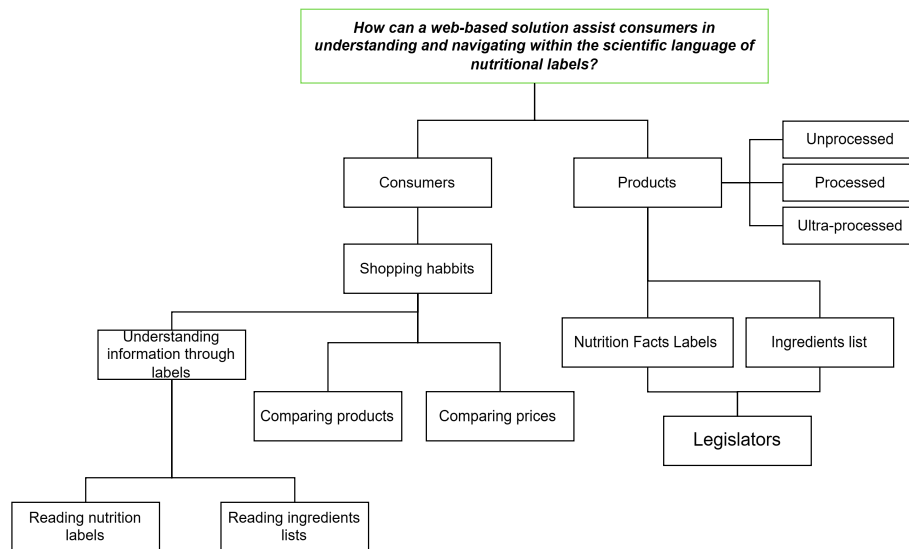


Figure 3.3: Decomposition chart

We have identified the product in and of itself, along with the consumer of said product to be our main components within our problem statement. Based on the data, we found, we argue that their ability to understand nutrition labels influenced their shopping habits. Additionally it has influenced their ability to compare said product to others of the same kind. This is one of the main reason as to why the ingredient list and **NFT** on the products serve as the other main component. Besides this, we also see that the level of processed ingredients is a concern to the consumers, which we will expand upon later in chapter 3.4.

Pattern Recognition

Following the process of decomposition, we seek to find any patterns in between each of the smaller problems. This step focuses on finding similar features across the different pieces, and helps to understand the bigger problem better. This revolves around finding similar properties across differing decomposed parts, which in turn will further our ability to solve the underlying complex problem, as stated in the problem statement (Csizmadia et al. 2015). Our pattern recognition is based on patterns seen in existing research and our own collected data of questionnaire and qualitative interviews. Thus, the figure below illustrates our pattern showing whether consumers understand nutrition labels of products or not, and how it affects consumer behaviour (see figure 3.4).

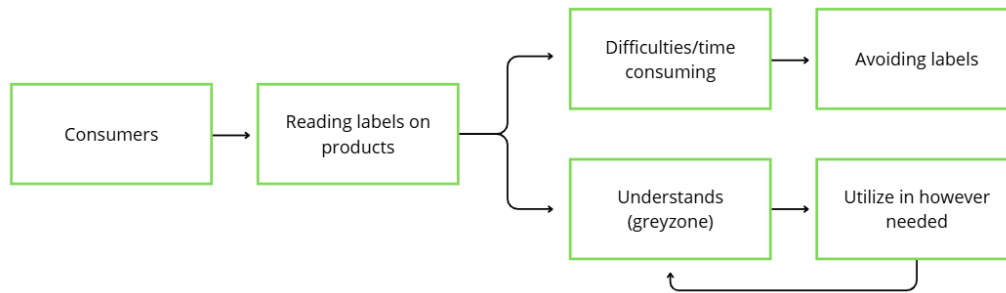


Figure 3.4: Pattern of user understanding

We found that the users will likely do one of two things depending on their ability to understand the information, given on the product. If the consumer does not have the ability, want, or time to fully understand the information, given through labels, they are likely to not repeat the action, and therefore not read the labels at all. On the other hand, we have found the pattern to change considerably, if the consumer *do* understand the information they read. This will instead form a reoccurring pattern, in which the consumer will read and potentially gain knowledge from the key information they seek, or perhaps the full information given by the product labels.

Clarifying these patterns within the problem area aids in understanding why the problem occurs, as well as why we find it important to study and create a web solution to support consumers in understanding and reading labels. Furthermore, these patterns, along with existing research and collected data, leads us to defining requirements both for users and the technical system, which we will explore in chapter 3.4.

Abstraction

The third component within CT is abstraction, in which we seek to remove any factors that does not directly involve or impact the solution to the issue at hand (Csizmadia et al. 2015). In this part we tried to come up with all the questions that came to mind, when thinking about our project and problem statement (see figure 3.5).

Pre Abstraction	Post Abstraction
Is a financial aspect important for our web solution?	How can label information become more efficient?
Do consumers compare products based on nutrition values?	Are consumers aware of different elements of nutrition labels?
How do consumers read nutrition labels and ingredients?	How can label information become more efficient?
How can label information become more efficient?	
Are consumers aware of different elements of nutrition labels?	
Does it matter who our target group is and what is the impact?	
Do consumers find information reliable?	
Do consumers find information reliable?	
Do consumers find nutrition labels/ingredient lists important?	
Do consumers' behaviour differ between reading labels on unprocessed and ultra-processed foods?	

Figure 3.5: Abstraction, pre to post

These questions have been written down in 'Pre Abstraction'. Some of the questions are relevant, while others are not, or are questions we will not have the opportunity to answer throughout this project. In 'Post Abstraction' we have filtered these questions from pre abstraction, down to the questions we find to be the core of the problem area. The final questions, presented in post abstraction, can be seen as the crux of this project, and are therefore necessary in order to answer the problem statement.

Algorithm Design

Lastly, algorithm design is a systematic approach that offers step-by-step instructions for solving problems by utilising decomposition, pattern recognition, and abstraction (Csizmadia et al. 2015). The algorithm design of this project will be presented in chapter 3.5. This decision, to postpone the algorithm design, is based on a need to firstly address the system requirements found within our data, as they are needed to fully define the design of our algorithm.

3.4 System Requirements

In order to best address the problem statement, we found it relevant to make a list of requirements, based upon the existing research and the collected data. In this section we will present said requirements, in which we will address their significance in respect to the users needs, as well as the functionality of the system.

MoSCoW Method

Requirement engineering is often an important step within the lifecycle of software development (Hudaib et al. 2018). Requirement engineering consists of several activities, however this chapter will focus on requirement prioritisation; a significant process within the work of a software project. One technique within this field, is the MoSCoW method, which is used to create a hierarchy of priorities, especially used in project management. This is a method created to help decide which requirements, tasks, or features to focus on. The acronym MoSCoW represents four categories of initiatives: *must-have*, *should-have*, *could-have*, and *won't-have* (Hudaib et al. 2018).

To effectively prioritise the requirements for our web solution, we utilised the MoSCoW method. This specific method was chosen, as it provides a clear framework for distinguishing between the critical features and those that are desirable but not essential for the initial version of the product. By categorising the requirements, we ensure that the essential functionality is present, as well as also considering enhancements for future work (Hudaib et al. 2018).

ID	Requirements	Implementation	Priority	Source
R1	The ability to view macro-nutrition	Macro-nutrition will be viewable on the product page.	Must have	<u>Existing research</u> : Shamim, Khan & Ahmad, 2020. <u>Questionnaire Q5-6, 9-10</u> : Mentions of interest in reading and prioritising macro-nutrition
R2	The ability to view micro-nutrition	Micro-nutrition will be viewable on the product page.	Must have	<u>Existing research</u> : Shamim, Khan & Ahmad, 2020. <u>Questionnaire Q5-6, 9-10</u> : Mentions of interest in reading and prioritising micro-nutrition
R3	Data should be easy to understand	Minimal information and translated to make it understandable.	Should have	<u>Existing research</u> : Europa, 2024. Shamim, Khan & Ahmad, 2020. Acton et al., 2016. Hammond, et al. 2023. <u>Questionnaire Q14</u> : Indication of difficulties understanding labels and feeling overwhelmed.
R4	Basis of comparison with other products	On product pages should be the ability to see other products related or as substitution.	Should have	<u>Interview 2, line no.</u> : 200-203 <u>Interview 2, line no.</u> : 209-210
R5	Have an interesting visualisation of product information	Colour code e.g. like traffic lights or smileys in green, yellow, and red.	Could have	<u>Existing research</u> : Acton et al., 2016. Hammond, et al. 2023. <u>Interview 3, line no.</u> : 315-320
R6	Accurate/trustworthy information and content	Users should see all information, and be able to read more if interested. Transparency on product pages.	Must have	<u>Existing research</u> : Europa, 2024. Campos, Doxey & Hammond, 2011. <u>Questionnaire Q6</u> : Mention of honest appearance, quality, minimal chemical.
R7	Time saving, fast and simple information	Easy access to search bar and information on webpage.	Should have	<u>Existing research</u> : Shamim, Khan & Ahmad, 2020. <u>Questionnaire Q14</u> : Demand on quick ability to select foods based on nutrition information. <u>Interview 5, line no.</u> : 424-428
R8	Financial aspect to save money and making grocery shopping more economical	A comparison of prices on products in different stores and between similar products.	Could have	<u>Questionnaire Q5-8</u> : Demand and experiences with restrictions based on economical situation.
R9	Information on both nutrition labels and ingredients	On product pages both content details are visible and explained.	Must have	<u>Existing research</u> : Europa, 2024. Acton, et al. 2016. <u>Questionnaire Q11-14</u> : Indication of a somewhat interest in both nutrition and ingredients.
R10	Give more information on health related to specific ingredients	Ingredients known publicly to be unhealthy in any amount should be marked to differentiate.	Must have	<u>Existing research</u> : Monteiro, et al. 2018. Europa, 2024. <u>Interview 1, line no.</u> : 96-98 <u>Interview 2, line no.</u> : 180-181 <u>Interview 3, line no.</u> : 327-330

R11	The user can prioritise which information they find most useful or necessary for specific health situations	There should be a filtering list option for users	Should have	<u>Existing research:</u> Persoskie, Hennessy & Nelson, 2017. Shamim, Khan & Ahmed, 2020. <u>Questionnaire Q11-13, 17:</u> Different interests in protein, carbohydrates, fats, etc. <u>Interview 4, line no.:</u> 370-372
R12	Information to differentiate between (ultra)processed foods and unprocessed foods	On the product page it should be visible what level of processed the food is.	Should have	<u>Existing research:</u> Europa, 2024. Monteiro, et al. 2018. <u>Questionnaire Q6, 9-10, 11-13:</u> Mentions of nutritional foods, vegetables, meat, fruit, food pyramid etc. <u>Interview 2, line no.:</u> 243-249

Figure 3.6: MoSCoW based on requirements

Prioritisation, as shown in figure 3.6, was determined by user feedback obtained from questionnaires, interviews, and an analysis of existing research. Each of the requirements' placement within the MoSCoW model reflects its importance to user needs and its practicability to the project. To further indicate the necessity of the individual criteria, we colour coordinated the 12 identified requirements. The category of won't-haves are not included in the figure above, but will be expanded upon in chapter 6.1.3.

Expanding upon the requirements

The following section will further elaborate on our requirements. To ensure the functionality of our web solution, we identified a set of requirements that will be explored in greater depth. These requirements form the foundation of the web solution, addressing critical aspects of user needs and aligning with the primary objective of enabling informed decision-making. By expanding on these specific requirements, we aim to clarify their significance and outline how they contribute to the overall functionality and user experience of the solution. Through the process of identifying requirements within our empirical data, we found that there are multiple key functions that must be upheld to secure functionality, that will sufficiently inform the user of the nutrition content and ingredients of a product (*R9*). This requirement in particular has been designated as a must-have, as providing in depth information to the consumer is the core purpose of the web solution. Without it, the solution fails to meet the primary objective, of enabling the user to make informed decisions. Additionally, we identified the importance of providing users with the ability to differentiate between ultra-processed and unprocessed foods (*R12*). This is particularly valuable to users who prioritise natural or minimally processed options in their dietary choices.

As a should-have requirement, the implementation could involve hover text or clear visual indicators to flag processed versus real foods, helping users make more informed decisions. Feedback from interviews highlighted concerns over artificial additives and the lack of clarity around food authenticity, reinforcing the relevance of this feature. Similarly, we found that users expressed a strong preference for transparent labelling of ingredients that are known to have health risks (*R10*). This must-have requirement would involve highlighting potentially problematic ingredients, such as added sugars or trans fats, with distinct visual cues

to ensure users can easily identify these. As users frequently voiced concerns about the health implications of consuming overly processed or harmful additives, this feature ensures that the solution remains health-focused and supports informed decision-making. In figure 3.6, we have decided to include the column 'How to implement'. This is the earliest work of how we have imagined the design and functionality to take place within the web solution. An example of this can be seen within *R1* and *R2*, our two other must-have requirements. These account for the ability to view both micro- and macro nutrition of the product, which should be displayed openly on a front page whenever the user searches for a product.

The emphasis on nutrient transparency, reflects a recurring theme in user feedback, highlighting the importance of accessible data for informed choices. Finally, we also included the aspect of the solution to be fast and simple as a must-have requirement, mainly given the 'How to implement' aspect of *R7*. Here we identify having a search functionality or the ability to scan a product to be of the utmost importance, as otherwise, the user would not have a direct way to access the information about a specific product. Furthermore, we have included several should-have requirements, that seek to elevate the experience, where must-have requirements are the key for the web solution to have any resemblance of functionality. This is where *R3* and *R5* especially becomes relevant, as we seek to make the information as easily accessible as possible for the user, where both our interviews as well as existing studies agree on the interesting visuals being a grand part of easy understanding (Hammond et al. 2023). Such tools align with the feedback from interviews, where users expressed a preference for visually engaging designs that facilitate quicker decision-making.

In developing the web solution, we also considered the limited time frame, as well as group members coding experience, leading to some features, such as economic aspects as well as comparison tools, being deferred to future iterations. These requirements will not be fully excluded, but rather postponed, in order to focus on the core functionality and ensure a timely delivery of the product. We argue, that by iteratively addressing further requirements in later stages, the web solution has possibilities of further growth to more comprehensively meet the users' needs.

3.5 Flowchart

As previously stated, now that the system requirements have been defined, the final stage of **CT** involves developing step-by-step instructions for how the system should function. These instructions can be interpreted differently by various individuals, depending on their perspective; however, in the case of this algorithm, they must be clear and systematic to ensure seamless translation into a computer's binary logic (see figure 3.7).

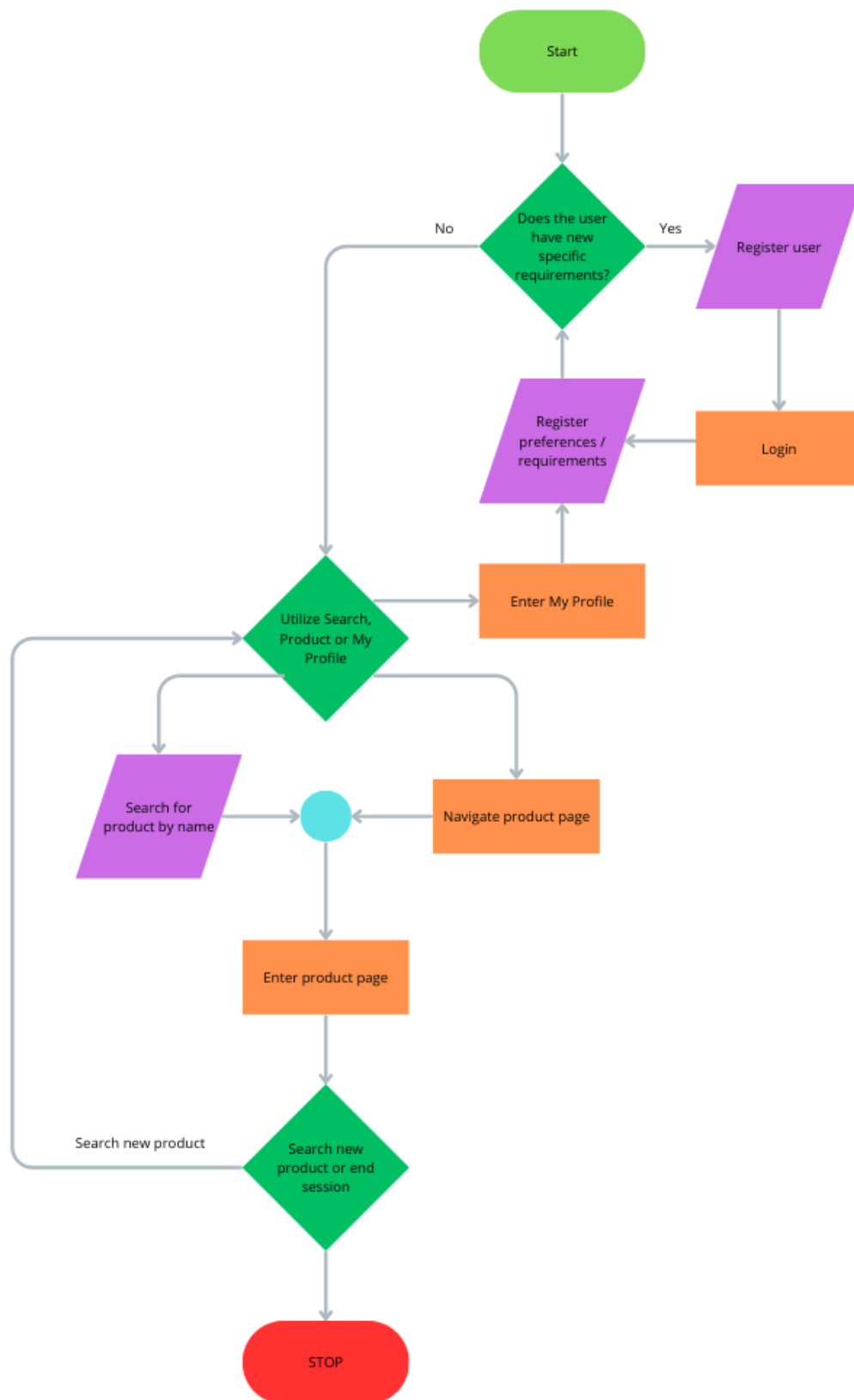


Figure 3.7: Flowchart

Working with the flowchart above, has been an iterative process, in which we first wish to introduce a component that has been added in alter stages of the process. Thus, the current version of the flowchart, reflects a close to final iteration of our product.

Firstly, we find it necessary for our web solution to have use cases, both with specific and generic needs. Therefore, we apply the first decision of the system, of whether or not, the user has a need to specify preferences for the information they want to gather. To do so, will require a login function so the choices in preference can be applied onto every product that the user would either scan or search for. This loop will repeat, until the user can confirm, that they have no new preferences that they would like to keep track of. This will then lead to the next decision, where the user is presented with the home screen, where they should have the ability to either re-enter their profile site to change their preferences, or proceed to searching for any given product, either through the manual product page, navigating through categories of items, or by searching for the product name through a search bar. The choice to navigate through the manual product list might seem slower and counter intuitive at first, but we argue that this gives the user a greater ability to look up products of the same sort that they are currently searching, which serves as an indirect way to compare different products within one category. We find this to be a way to achieve one aspect of the comparison tools we identified to be wanted from our interviews and questionnaire.

Finally, the last decision of this algorithm asks if our user has further use of the web solution for the given moment, which will then either repeat the previous steps, or bring the algorithm to a halt.

4 Finding and exploring the target audience

The following section will introduce and explore the target audience for our web solution. This is an important part of succeeding in effective communication, as it helps in making the message more relevant and effective on the recipient. To be able to identify and segmentate the target audience it is crucial to understand audience characteristics and decide on a primary group (Lin 2002). However, it can still be somewhat relevant to reflect upon other secondary groups as they can assist in spreading awareness of the product. In this project we focus on a primary group of young adults between 15-30 years old, but reflecting upon which secondary groups could be used to motivate the younger generations to use our web solution could quite possibly be older adults, most likely parents, or internet influencers, given the effective power of media and trends. This consideration is based on the fact that there are quite a lot of research and social learning theories, that suggest that the social environment, support, norms, peers, among other play an important role in behavioural patterns of the individual (Dhiman 2023).

To narrow down a certain target audience, it is important to know what the main problem is and what the audience's characteristics and behaviour are. For that reason, our defining and exploring of the target audience will be based on the problem analysis and data we collected. Ph.D.. in Business Administration Chin-Feng Lin distinguishes between demographic segmentation, such as age, gender, status, culture, and psychographic segmentation, such as belief, values, lifestyles, and personality traits. Furthermore, Lin suggests how a combination of these two approaches helps in identifying a clear target audience and more focused communicative strategies (Lin 2002). The first part of identifying and exploring our target audience, will be focused on demographic factors. As mentioned in the methodology chapter (see chapter 3) our aim has been young adults between the age of 15-30 years old. We have not been focusing on the gender of the individuals as we asses that learning to use and understand nutrition labels should not be limited to specific genders, however as the existing research presented in the problem analysis suggests, there seems to be a slight difference between what nutritional values interests females versus males (see chapter 2.2). Furthermore, the research mentioned that consumers with longer education tend to understand nutrition labels better than consumers with a high school diploma or less (see chapter 2.2). With this in mind, the demographic factors for our target audience have been quite broad, as the primary focus has been on both men and women aged 15-30 years old and of any education level. This age-group was chosen based on research showing that they are more affected and vulnerable to the issue, as their consumption of ultra-processed foods were higher, making them a relevant focus for this project (Hammond et al. 2023). Our target audience's age group was as mentioned 15-30 years old, which is where the individual transforms from adolescence to young adulthood. This is a period, where the individual experience more independency, identity development, cognitive development, but are still influenced heavily by their social group, family, education system, workplace, among others (Vorhies et al. 2009). This is another reason, why we find this

target audience relevant, as it indicates that it is in this age group, that the individual develop and establish more adult-like preferences and behaviours, so encouraging an interest in reading and understanding nutrition labels seems fitting.

To explore our target audience even further, we also focused on psychographic factors such as lifestyle, values, and personality traits according to nutrition labels and grocery shopping. As mentioned in the problem analysis, the existing research shows a tendency that certain people struggle to understand and use nutrition labels, which some studies found to be because of the time effort and difficulties in understanding terminology. This problem was also somewhat mentioned in our data, and is expressed in the following quote *“Processed foods have so many chemicals and ingredients that it feels too complicated to understand how they affect health”* (appx. 5, line 392-396). This could indicate the need for simpler labels and better education to help people make healthier informed choices. This also comes to light in the following: *“I don’t have much knowledge about what nutritional information actually tells me, but I find the ingredient list very interesting and relevant. I like to see if real ingredients are used, not something highly processed or artificial, like E-numbers”* (appx. 5, line 240-246). Some people find it difficult to understand labels, but nonetheless still have some curiosity to whether the food or ingredients are healthy or not. This concern of not understanding what different content actually means, is also expressed in the following quote: *“I think it’s fine as it is. The problem is more that people aren’t informed enough about what the different nutrients do to the body.”* (appx. 5, line 384-386). Here the respondent expressed that they don’t have the knowledge of what the different ingredients do to their body, which potentially can lead to a serious issue for those who have allergies or health conditions, where diet plays a crucial role, such as diabetes or high cholesterol. This was also argued in the existing research and data on nutrition (see chapter 2). Furthermore, another quote indicated that the target audience might rely on the nutrition laws made by legislators such as EU, and Denmark in this case, where the respondent mentions *“As a starting point, I trust the initiatives taken at both EU and national levels regarding what can and cannot be included in food products”* (appx. 5, line 34-40). The respondent trusts the EU’s food legislation, which aims to protect human health and ensure the proper functioning of the food market. The EU’s food safety policy covers food “from farm to fork” and guarantees safe and nutritious food along with high standards for animal health and welfare (Schebesta, Bernaz, and Macchi 2021). While this seems to bring comfort, it is still important for the individual to take account of their daily intake and personal diets. Understanding the target audience is crucial to being able to design an effective web solution, we first identified the needs and requirements of our target audience: we found our target audience to be young adults aged 15-30, seeking quick, accurate, and accessible nutritional information to be able to make healthier lifestyle choices. Through research and analysis, we found the key priorities to be usability, time-efficient search functionality, the need for clearer, more accessible information, as well as visually engaging design elements. These findings was the foundation for our decision to create NutriScan, a web solution that combines simplicity, interactivity, and transparency to meet these expectations. In the following chapter, we will go into detail on how we focused on addressing the audience’s needs by leveraging HTML for structure, CSS for aesthetics, and JavaScript for interactivity.

5 Implementation

This chapter will present the structure of the design process and the tools and technologies that are incorporated into the workflow for developing the webpage's prototypes. Additionally, it provides an overview of the programming languages and code used during the development process. All code snippets presented in the implementation can be found in a larger image in appendix 1.

5.1 Wireframe of low-fidelity prototype

Once the system requirements and algorithm design are defined, the next natural step is to begin prototyping. As an initiating process to this, we started by making a low-fidelity (lo-fi) prototype in the program Figma. Digitally sketching out our ideas for the prototype worked well, as it made it possible to form different shapes and illustrations, and made it easy to move them around or delete if not needed. This helped in giving enriching and reflective dialogues among group members on different layout ideas and how the requirements should fit. Furthermore, we found it effective to use a digital platform like Figma as all members would be able to access the lo-fi prototype whenever and wherever needed.

Prototyping is where we go from the concept we created through the steps of CT to development. During this it is possible for all members of the group to participate in brainstorming, creativity and experience with different functionalities, interactions and user-flow (Lee 2020). Furthermore, making prototypes also helps in realising errors earlier in the process, which helps the overall process, as Ph.D. in Design and Technology Yu Hin Brian Lee cites "*Fail earlier to succeed sooner*" (Lee 2020, page 55). We did this, to try and figure out how we could incorporate the requirements that we found to be essential to the application, such as those regarding display of information in an orderly manner, as well as granting the user further ability to gain deeper insight into said information (See R1/2 & R5/6 in figure 3.6). Our lo-fi prototype can be seen in figure 5.1 below. It illustrates our initiating ideas and visions for the webpage, such as the colours we imagined would work well with the overall theme of nutrition, as well as our requirements, how we wanted the structure of the webpage, what content should be shown and accessible, amongst other things.

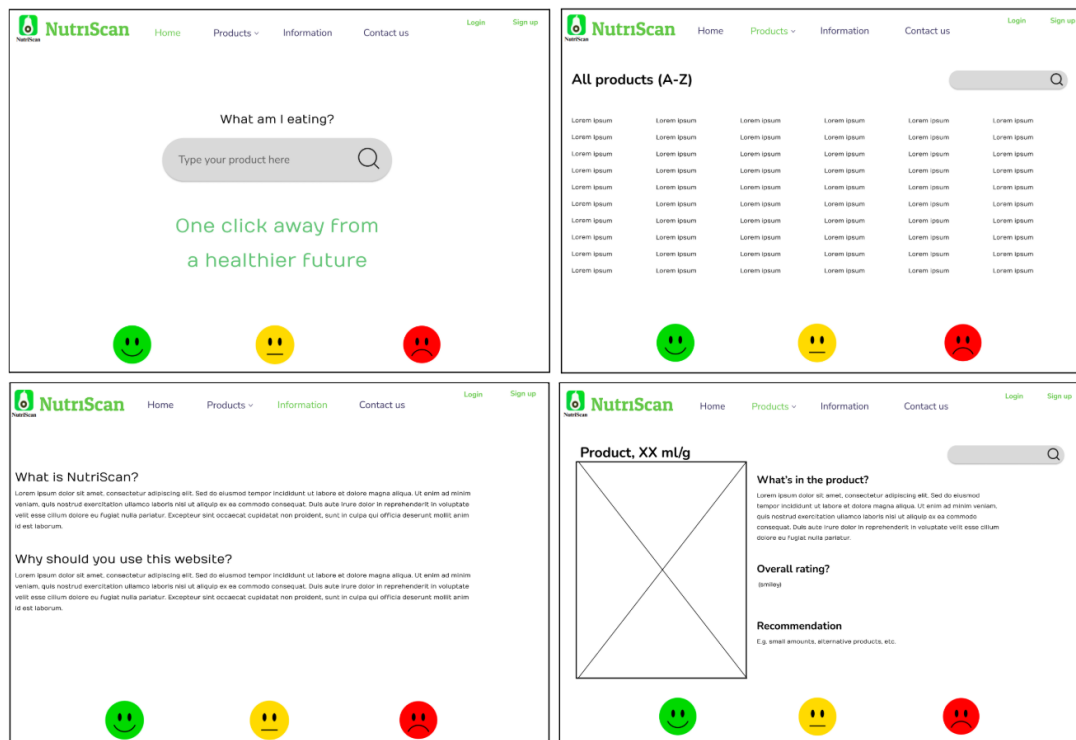


Figure 5.1: Low-fidelity prototype in Figma

To make our prototype more understandable the following part will include wireframe description. This is typically where interface design, information and navigation design is connected and presented. This is in other words a visualisation and description of the webpage's skeleton. However, it is not always necessary to include all pages in the wireframe if the webpage's build is less complex, which is also why we, during our system description, mostly focus on the home page, product page and the user profile (will be introduced in 5.2), which carries out the functionalities. When creating the visual and functional design of the web page, the wireframe is an imperative step as it outlines information structure, interactive and visual elements and it is guaranteed that when it is actually time to develop the system this wireframe, initiating sketches, and lo-fi prototypes will become helpful (Garrett 2011).

In the top of our prototype, is the global navigation, which from the left corner includes a logo to help identifying our web solution, and by clicking on the logo the user will be taken back to the home page. Following the logo appears the main navigation menu, which includes different pages that the user can access on the webpage. As our lo-fi prototype illustrates we have already made some ideas as to which pages, such as a drop-down menu for products, information, and contact. The last part of the global navigation is the login and sign-up, which should evoke a call to action and motivate the user to integrate the webpage more in their daily activities, such as grocery shopping, by creating a profile. On the homepage the first content to appear is the header, which purpose again is some sort of call to action to motivate user engagement. As seen in figure 5.1 we wrote the line "What am I eating?" to hopefully spark an interest or reflection in the user. Directly underneath is a central and enlarged search bar, which we intent to make conspicuous as we

wish for the user to utilise this function. Lastly, on the homepage is a supportive catchphrase to bring some sort of value to the search tool, this can also be seen in figure 5.1 written as “One click away from a healthier future”. In addition, the footer illustrates three smiley icons coloured green, yellow, and red which displays or hopefully connotes to a rating system, as we delved into in chapter 2.2.

The product page is displayed for the user, when a specific product has been searched or found through categories. The page includes a large picture of the given product, product title, and volume (ml/g) for recognition and easier identification. Beneath, the user is first introduced to the nutrition and ingredient details in an explanatory summary of the product content. After this, is an overall rating of the product, which initially was based on the earlier mentioned smiley icons, and lastly recommendation for healthier alternatives or how often the given product should be consumed. The wireframe is also visualised in figure 5.2 below.

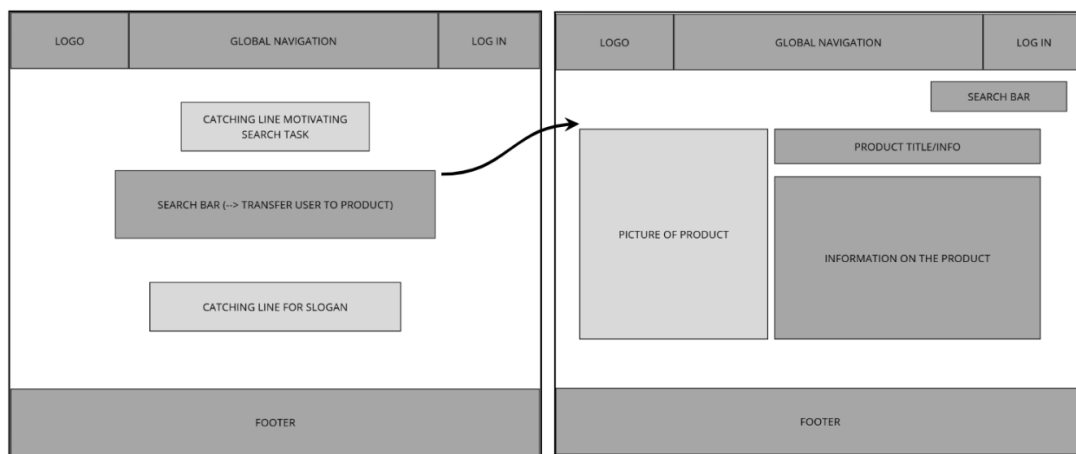


Figure 5.2: Visual wireframe

Because of the limited time given in the project assignment, we moved directly from this **lo-fi** prototype to a high-fidelity (**hi-fi**) prototype, which will be further explained in the following section, 5.2. Although, if more time were given, it would have been convenient to spend more time on this **lo-fi** prototype and iterating ideas and feedback. Furthermore it would have been beneficial to make a medium-fidelity prototype before the actual coding. Figma does not only give the user the ability to create a sketch of a prototype, but it is also possible to create basic UI elements, such as working interactive elements and functionalities, however without us having to code anything. With a few buttons it is possible to create an almost realistic looking webpage, which could have been effective for testing out functionalities and usability, and thereafter revise the design. To do such a testing before creating the **hi-fi** prototype would most likely have helped in saving time on multiple changes and uncertainty in the programming process.

5.2 Wireframe of high-fidelity prototype

The following will give an insight into the **hi-fi** prototype, the final prototype of this rapport. We will further explain the webpage's structure, visual elements and how the main functionalities work. The visual appearance of the web solution remains rather similar to the **lo-fi** prototype, with green as the overall theme colour and information layout. The final product can be seen in figure 5.3, and can also be found in appendix 1 for a larger version.

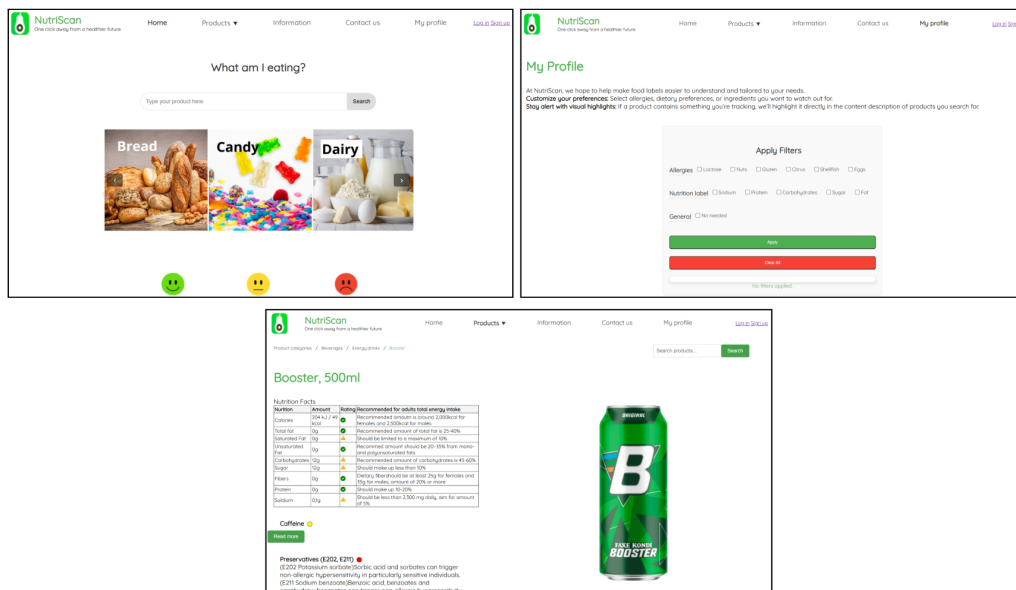


Figure 5.3: High-fidelity prototype (Visual Studio Code)

The home page has remained quite close to the original, however the slogan to catch the user's attention and briefly explain the goal or intention of the web solution, has been moved to directly beneath the name 'NutriScan' to support the name. Additionally, the home page now includes a picture carousel beneath the search function. The carousel displays different food categories as an alternative way to accommodate user needs. Using the carousel tool makes it easier to display multiple pictures at a time, while it scrolls through the pictures automatically, however the user is able to use the arrows to scroll to either left or right as well. The functionality is also that the user can access needed categories directly by clicking on the wanted category. Our intention with the carousel is also to motivate the user to scroll through the pictures, a desire to explore our web solution and become aware of other food categories, they may not have thought about earlier. Besides this, it also helps to give a more visual element to the front page, which has a more simple appearance. The simple minimalistic design is however intentional as well, as we want it to be an 'easy access' tool for the user, potentially while being at the grocery store. We assessed that a lot of pictures, information and pop-ups could risk confusing or overstimulating the user, and is also not suiting the problems argued in the problem analysis, that some people tend to find nutrition information and labels confusing and time consuming (see chapter 2). A combination of a minimalistic visual appearance and the use of short sentences on the web solution, as well as the use of transparent categories helps in limiting cognitive load for

the user, as they do not have to process a lot of information, colours, pop-ups, etc. This is also quite fitting the web solution, which purpose is to convey data, information or to include user tasks (Tidwell, Brewer, and Valencia 2020) (Tidwell, Brewer, and Valencia 2020).

The product page with information of a chosen product has also remained in the original idea, though the layout and information has been changed. The user will be able to view a **NFT**, as they know from the actual physical product, however we have added a short description of the total amount recommended for adults for each type of nutrition. In addition to this, there is a simple rating, which indicates whether the nutrition amount is appropriate or concerning in relation to these recommendations. This can be seen in the previous figure 5.3. Beneath the table are the listed ingredients, with a read more/less functionality. This makes it possible for the user to click on certain ingredients and get more information. These are also complemented with the ‘traffic light’ rating, but compared to the **lo-fi** prototype, there are more functionality to the rating and information now. We chose to keep some sort of rating system as the existing research argued that the ‘traffic light’ is more intuitive, likewise did some of our respondents point a preference for easy colour indicating (appx. 5, line 315-319) and parameters to compare with (appx. 5, line 209-210).

Lastly, the new additional functionality, added to the prototype is the profile page, which purpose is to give the user a more personalised experience when using the web solution. The user is able to select specific ingredients and nutritions based on allergies, lifestyles choices and more. If the user chooses to apply these filters, our intention is, that the selected ones will be highlighted on product pages if, they are included in the products content. Our intention with this is to make the information of content more comparable for the user. By doing this, it will hopefully be easier to compare products and quickly recognise products with either generally ‘unhealthy’ ingredients or unhealthy to specific allergies or diets. In addition to this, making the content more personalised should also foster more user engagement by custom filters and therefore give more motivation and significance for the user. We also assess that this functionality is where our web solution stands out from existing alternatives.

The two main functionalities, or user journeys, with the web solution is visualised in figure 5.4. These are categorised into the search function and get information on a specific product, and to sign up or login to receive more personalised information. For this reason, these two functionalities are colour highlighted in figure 5.4 below. It is relevant to point out the fact, that for these functionalities to make an impact, they have to be driven by a user motivation. As argued earlier e.g. when defining the target group, it was mentioned that our quantitative and qualitative data indicates an interest in learning to understand nutrition labels. This motivation is needed in order to bring the web solution purpose.

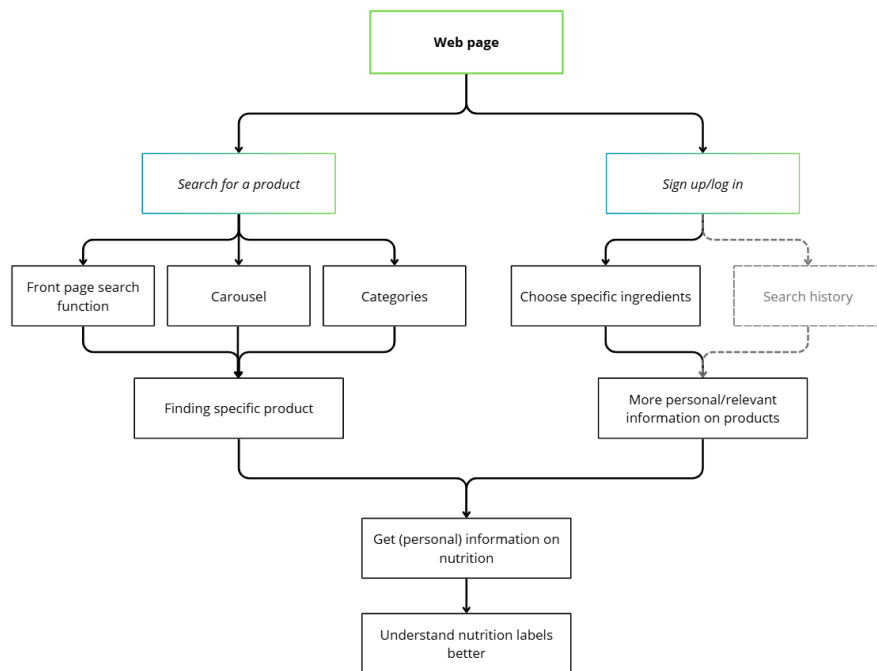


Figure 5.4: System Overview

The figure likewise illustrates the structure of the web solution and the different elements we implemented. The workflow and source code used, will be explained in the following chapters. As visible on the figure, we wanted to add a feature of a search history on the profile page, where the individual user's previously searched products would be listed or appear. However, we did not manage to incorporate it due to limited time, but the intention with a search history functionality is to make it easier for the user to retrieve earlier visited products, to view the information again or to read more into it. This function could for instance become convenient in order to revisit the products after returning from the grocery store to further delve into the product content.

Another way to visualise and describe our web solution is that we make use of the navigation model *Multilevel pattern*. This can be seen in figure 5.5, where the main pages of the header all connects to each other, however the subpages under products only connect to the products page and through the carousel on the front page. These pages only connect among themselves, but users can always navigate back to the main pages through global navigation (Tidwell, Brewer, and Valencia 2020). That being said, the user will need to make a few steps on the webpage to get to a specific product.

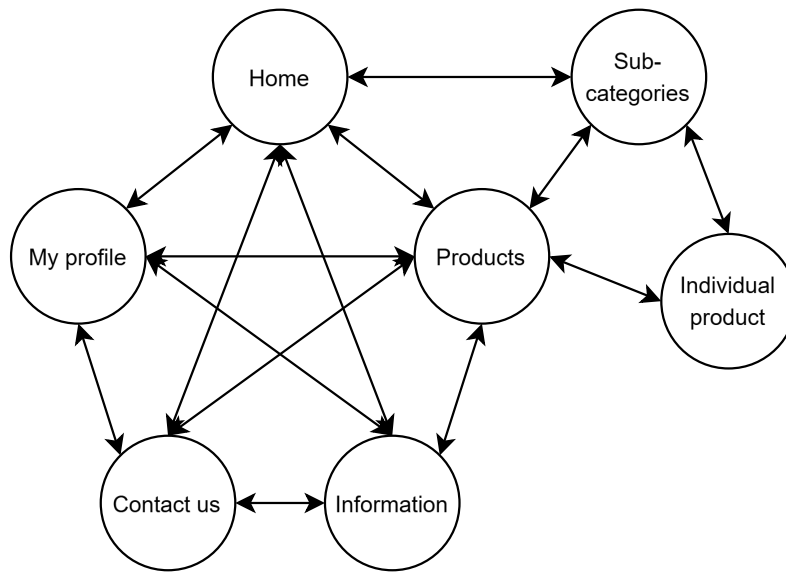


Figure 5.5: Multilevel Pattern

To make the navigation more user friendly we implemented a breadcrumb element on the products pages, to visualise the navigation hierarchy. The purpose of this, is to give a better user experience as the breadcrumb works similar to a "you are here"-sign and gives the user an understanding of what steps they took, to reach their current page and how to navigate back (Tidwell, Brewer, and Valencia 2020). An example of breadcrumb in our web solution can be seen in the figure below, 5.6.

Product categories / Dairy / Cheese

Figure 5.6: Breadcrumb

5.3 Workflow

To enhance the efficiency of our workflow, we applied different programs and tools such as pair programming, source code management (GitHub), and an integrated development environment (Visual Studio Code). We used GitHub, an online development platform and cloud-based service, where you can store, share and manage the repository, for our project, facilitating collaboration during the coding process (GitHub 2024). Additionally, GitHub made our repository available to all group members. The branching features that are provided for repositories in GitHub, permitted team members to work on different components of the program simultaneously. This method was primarily used to set up the basic structures of our source code. Afterwards, the individual branches were merged into the main branch program, to integrate the collective contributions.

Our code is developed using the source code editor Visual Studio Code, making it possible to access the GitHub repository and edit our source code (Visual Studio Code 2024a). The reasoning behind this, is that this editor is compatible with almost every programming language. It also contains built-in debugging tools, which simplifies the process of identifying and resolving potential bugs in the code (Visual Studio Code 2024b). This web solution is primarily built using a software development technique known as pair programming, after establishing some of the basic structures of our code. Pair programming is said to be “*an Agile software development technique originating from Extreme programming (XP) in which two developers team together on one computer*” (Gillis 2021). In our case, we used an adapted form of pair programming, by using the Live Share extension in Visual Studio Code. This tool facilitates seamless real-time collaboration among developers by executing tasks such as ‘push’ and ‘pull’ operations, allowing simultaneous coding, reviewing, and discussion within a shared workspace (Silver 2018). This method helps teams catch issues early and makes it easy to collaborate more efficiently (Steck 2023).

5.4 Used Programming Languages

In the following section, we provide a brief overview and explanation of the programming languages and frameworks used in the development and design of our web solution. Our prototype is built using a combination of tools and technologies, including the web track from CS50, which includes instructional videos on [HTML](#), [CSS](#), and [JavaScript](#). These resources have been contributory in managing the structure, style, and functionality of the prototype.

HyperText Markup Language [HTML](#)

[HTML](#) is a language that serves as the fundamental building block of the Web, as it provides the structural foundation for webpages. It defines the meaning and arrangement of web content, hence allowing developers to organise text, images, as well as other elements using markup tags. Each [HTML](#) element consists of an opening tag - the content - and a closing tag, with attributes that can be assigned to the element’s properties or connect to styling information. [HTML](#) is a markup language, rather than a programming language, it is primarily used for structuring web content along with [JavaScript](#) and [CSS](#), where [JavaScript](#) is in charge of the functionality and behaviour, and [CSS](#) is generally used to describe a webpage’s appearance and presentation. HyperText is a feature of [HTML](#) that refers to the use of links that connect webpages either within the same webpage or across different sites, simplifying interactions and navigation (MDN Web Docs 2024b).

JavaScript (JS)

JavaScript is a lightweight interpreted programming language, used to create dynamic and interactive features on webpages. JavaScript is a prototype-based, multi-paradigm, dynamic language that supports object-oriented and declarative styles amongst others. Unlike [HTML](#), which defines the structure, and [CSS](#) which manages the style, JavaScript allows interactivity and behaviour ([MDN Web Docs 2024c](#)). We used JavaScript as a complement to our [HTML](#) code, as [HTML](#) alone cannot react to user actions. Therefore by combining JavaScript and [HTML](#), we allow the webpage to respond to events like these, hence bringing these elements to life.

Cascading Style Sheets CSS

CSS is like [HTML](#), not a programming language, however, it is a style sheet language and is used to control the presentation and layout of [HTML](#) elements on a webpage ([MDN Web Docs 2024a](#)). CSS describes how elements should be displayed on screen, on paper, in speech and more. While [HTML](#) provides the structure and content, [CSS](#) defines the visual appearance. This separation between content and design makes it easier to organise during the web development process.

5.5 Source Code

The following section describes the development and functionality of the NutriScan web solution, highlighting key features and their underlying code. The web solution is built with a combination of [HTML](#), [CSS](#), and JavaScript, and is designed to provide users with an easier and faster experience, for searching and understanding nutritional information. The code snippets below show how various aspects of the code were implemented, focusing on functionality, layout, and design. The code snippets below will be presented in the following order: The interface, including the different subpages of our application. Then the product details pages, with a focus on the display of the nutritional values and health indicators. Afterwards, the layout and design choices.

5.5.1 Homepage

The interface includes the main components of the NutriScan web solution, this includes the *header*, *navigation menu*, *search bar*, and *main content section*. These elements are included to crease more usability, that implements interactions.

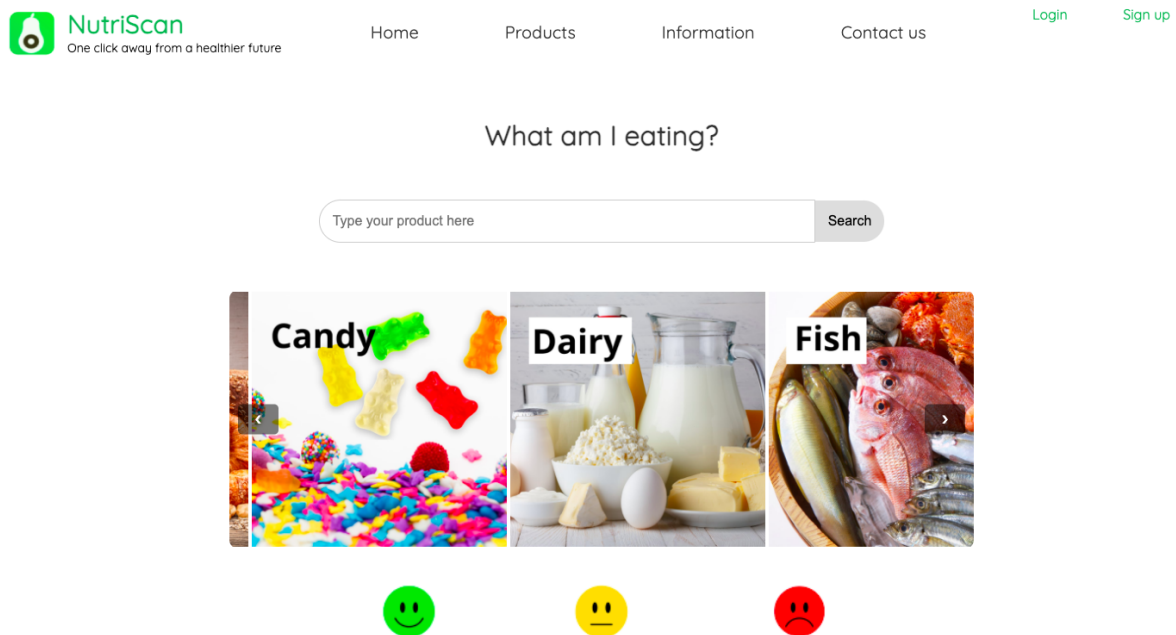


Figure 5.7: Homepage, NutriScan web solution

Header Section

The `<header>` element in **HTML** represents a webpage's top section and has the `id` of `header-section`, which is targeted in the **CSS** code for styling. In this case, it displays the NutriScan logo and navigation menu. The drop-down menu under *Products* allows users to view the available categories on the prototype, while the search bar provides quick and easy access to product information. The **HTML** code for the header section is shown in a smaller image in Figure 5.8 below.

```

11 <!-- Header Section -->
12 <header id="header-section">
13   <div class="logo">
14     
15   </div>
16   <h1 onclick="goToHomepage()">NutriScan</h1>
17   <p onclick="goToHomepage()">One click away from a healthier future</p>
18 </div>
19 </div>
20 <nav>
21   <a href="homepage.html">Home</a>
22   <div class="dropdown">
23     <a href="products_all.html">Products</a>
24     <div class="dropdown-content">
25       <a href="product_1.html">Redbull</a>
26       <a href="product_2.html">Shredded pizza topping</a>
27       <a href="product_3.html">Product 3</a>
28     </div>
29   </div>
30   <a href="information.html">Information</a>
31   <a href="contactus.html">Contact us</a>
32 </nav>
33 <div class="auth-links">
34   <a href="#">Login</a>
35   <a href="#">Sign up</a>
36 </div>
37 </header>

```

Figure 5.8: Header Section, HTML code

Inside the header section, the `<div class="logo">` contains the NutriScan logo as an image `<img>` and the webpage name as a heading `<h1>`. The title "NutriScan" and slogan "One click away from a healthier future" are clickable. When either is clicked, the `<goToHomepage()>` function is triggered, which is defined in the JavaScript code. This function redirects the user to the homepage by setting `window.location.href` to "homepage.html". The `<onclick>` function on the `<h1>` and `<p>` tags connects the elements to the `goToHomepage()` function, as seen in Figure 5.9 below.

```

13
14 //Function to redirect to the homepage
15 function goToHomepage() {
16   window.location.href = "homepage.html";
17 }
18

```

Figure 5.9: Redirect to the homepage, JavaScript

The `goToHomepage()` function changes the browser's location by setting `window.location.href` to `homepage.html`, redirecting the user to the homepage and improving user navigation and interactivity.

Navigation Bar

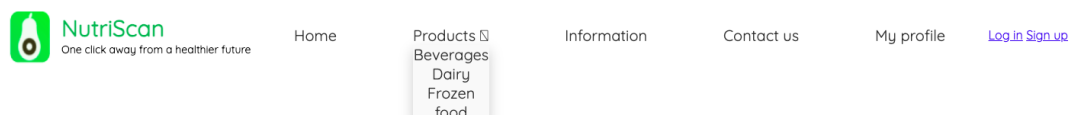


Figure 5.10: Navigation bar

Another feature implemented to improve interactivity and usability is the navigation bar seen in Figure 5.10 above. This includes primary links such as Home, Products, Information, Contact Us and My profile, ensuring that the user has easy access to key sections of the web solution. Additionally, the navigation bar incorporates a drop-down menu under Products, presenting links to specific category pages, such as 'Beverages' and 'Dairy'. As seen in Figure 5.10 the `<nav>` tag is used to group the navigation links (`<a>` elements) so that the user can move between different pages and subpages of the webpage. The `<a href>` elements within the `<nav>` tag define the clickable links, that direct the users to specific pages or subpages of the webpage. The `href` attribute specifies the exact file to which the user will be redirected to, upon clicking the link.

Search Functionality

The main function of the web solution is to allow the user to search for products and retrieve the products' nutritional information. This is illustrated on the homepage in Figure 5.7 with the search functionality in the middle, which allows the user to input product names and then be redirected to a product page, upon which they are given an overview of the nutritional information. This fulfils requirement *R7* (Time saving: fast and simple information) as well as *R1* and *R2* (Display nutritional values). After typing a product into the search bar and clicking the 'Search' button, the `searchProduct` function in JavaScript is executed due to the `onclick` event in the `HTML` code as seen in Figure 5.11 below.

```
43     <div class="search-bar">
44         <input type="text" id="search-bar" placeholder="Type your product here">
45         <button onclick="searchProduct()">Search</button>
46     </div>
```

Figure 5.11: Search bar, `HTML`

After executing the `searchProduct` function, the user's input is obtained from the search bar by reading the `HTML` element with the `id="search-bar"` using JavaScript.

The code `document.getElementById("search-bar").value` reads the text entered by the user in the input field of the search bar. To increase usability within the search bar, the search is case-insensitive so that the user can type the specific product in multiple ways, e.g. RedBull, redbull or in all caps REDBULL, this is possible because of the `.toLowerCase()`, which converts text to lowercase. JavaScript then handles that input in the `searchInput` variable and then it redirects the user to that specific product page, with the product's nutritional information, meeting requirement *R9* (Provide product information). Otherwise, an alert message notifies the user that the product was not found. Figure 5.11 and Figure 5.12 which can be seen below, show the `HTML` and JavaScript code for the search bar in a smaller image.

```
3 function searchProduct() {
4   const searchInput = document.getElementById("search-bar").value.toLowerCase();
5
6   if (searchInput === "redbull") {
7     // Redirect to the product
8     window.location.href = "products.html"; // Ensure the file path is correct
9   } else {
10    alert("Product not found!"); // Optional: Handle invalid searches
11  }
12 }
```

Figure 5.12: Search bar, JavaScript

The `<div>` element in the HTML code is a container that groups the search bar components, such as the input field and the button. The class name `search-bar` is assigned to the `<div>` using the `class`. This makes it possible for the group to be styled using CSS and targeted in JavaScript, adding interactive features that enhance the user experience. In the CSS code below in Figure 5.13. The flex properties in `.search-bar {` makes the input field and button horizontally aligned, and centred within the `.search-bar`.

```
151 .search-bar {
152   display: flex;
153   justify-content: center;
154   align-items: center;
155   margin: 55px auto;
156   width: 50%;
157 }
```

Figure 5.13: Search bar, CSS

Hereafter the property `align-items: center;` to make sure that the flex items are aligned vertically in the middle of the search bar. To make sure that the search bar is centred horizontally in the middle of the page, the left and the right margins are automatically calculated to centre the `.search-bar` as well as add 55 pixels of space from the elements placed above and below.

Carousel Element

Under the search bar, we have included a sliding carousel with navigation buttons, which is used to showcase a rotating set of images of the different food categories. Each image is clickable and redirects the user to the specific product category. This element is implemented to improve the user's engagement, by offering an interactive and visually appealing experience.

```

47 <div class="carousel">
48   <div class="carousel-images">
49     <a href="beverages.html"></a>
50     <a href="product_2.html"></a>
51     <a href="product_3.html"></a>
52     <a href="product_1.html"></a>
53     <a href="product_1.html"></a>
54     <a href="product_1.html"></a>
55     <a href="product_1.html"></a>
56   </div>
57   <button class="prev"></button>
58   <button class="next"></button>
59 </div>

```

Figure 5.14: Carousel element, HTML

```

30 function updateCarousel() {
31   const offset = -(currentIndex * (100 / imagesPerSlide));
32   imagesContainer.style.transform = `translateX(${offset}%)`;
33 }
34
35 //move to next set of images
36 function nextSlide() {
37   if (currentIndex < totalImages - imagesPerSlide) {
38     currentIndex++;
39   } else {
40     currentIndex = 0; // Loop back to the beginning
41   }
42   updateCarousel();
43 }
44
45 //move to the previous set of images
46 function prevSlide() {
47   if (currentIndex > 0) {
48     currentIndex--;
49   } else {
50     currentIndex = totalImages - imagesPerSlide;
51   }
52   updateCarousel();
53 }
54
55 // Attach event listeners to buttons
56 nextButton.addEventListener('click', nextSlide);
57 prevButton.addEventListener('click', prevSlide);
58
59 // Auto-slide every 5 seconds
60 setInterval(nextSlide, 5000);

```

Figure 5.15: Carousel element, JavaScript

As seen in Figure 5.14, which can be seen above, the carousel is grouped in **HTML** and styled as a cohesive unit in the main container `<div class="carousel">`. Within the main container, the `<div class="carousel-images">` serves as a flexbox container, that is used to arrange the different `<img>` elements in a horizontal row, which allows the images to have a smooth sliding transition. The user can move between the images by pressing the navigation buttons, which are prompted by `<button class="prev">` and `<button class="next">`. The JavaScript code which is seen in Figure 5.15 above, is directly connected to the **HTML** elements, to enable the interactions. The `updateCarousel` function calculates the correct offset percentage for the `<div class="carousel-images">` based on the `currentIndex`. The `nextSlide` and `prevSlide` functions are responsible for the logic of moving to the next or previous image. The functions send an update to the `currentIndex` based on the user's interaction and then loop back to the beginning of the carousel, when the limit of images is reached. After updating the index, the `updateCarousel` function is called to adjust the position. The event listeners attached to `<button class="prev">` and `<button class="next">` identify whether the user clicks and then triggers the relevant `nextSlide` or `prevSlide` function that is needed. To enhance the visual appearance, the `setInterval` function prompts the carousel to transition automatically every five seconds.

5.5.2 Product Details Page

The “Product Details” page as seen in Figure 5.16 below, provides the users with easy-to-understand information about the specific products’ nutritional values.

NutriScan
One click away from a healthier future

Home Products Information Contact us My profile Log In Sign Up

Product categories / Beverages / Energy drinks / **Booster**

Booster, XXml

Nutrition	Amount	Rating	Recommended for adults total energy intake
Calories	xxx	●	xxx
Total fat	xxx	●	Recommended amount of total fat is 25-40%
Saturated Fat	xxx	▲	Should be limited to a maximum of 10%
Unsaturated Fat	xxx	●	xxx
Carbohydrates	xxx	▲	Recommended amount of carbohydrates is 45-60%
Sugar	xxx	▲	Should make up less than 10%
Fibers	xxx	●	Dietary fibers should be at least 25g for females and 35g for males
Protein	xxx	●	Should make up 10-20%
Sodium	xxx	▲	xxx

Caffeine ●
[Read more](#)

Type of product, short quick product information ●
[Read more](#)

Type of product, short quick product information ●
[Read more](#)

Figure 5.16: Product Details Page

The product page is designed with the aim of providing users with easy and quick access to product information. The page features an intuitive structure, helping users gain a detailed understanding of the product’s nutritional values, ingredients and related information. At the top of the product page, there is a header section with a logo, navigation menu, and search function, as described earlier in the chapter 5.5.1.

The main section consists of a product title and an image of the product, which in this case is Faxe Kondi Booster, an energy drink. Both the image and the title are prominently displayed to help the user quickly recognize the product. The purpose is to convey the nutritional content of the product, which is presented in a table. The table provides a clear overview of the nutritional content of the product, such as calories, fats, carbohydrates, sugars, and proteins. To make the nutritional label easier to understand, colour-coded symbols have also been added. The green symbol represents the recommended values and the yellow symbol represents values that should be limited or be aware of. Below the table, interactive elements have been added in the form of ‘Read More’ buttons, placed under specific product information and sections. These buttons allow users to access additional details and explanations of various concepts. Additionally, colour-coded indicators are used to make it easy to assess the product’s status and health evaluation.

```

54     <body>
55     <main>
56     <body class="left-aligned-page">
57     <main id="main-section">
58     <h1>Booster, XXml</h1>
59     <div class="content">
60     

```

Figure 5.17: HTML - title / image

The **HTML** code is structured into the following key components as shown in 5.17 above. The `<body>` tag contains all the content of the webpage, while the `<main>` tag is inside the `<body>` tag, which serves as the container for the main content of the page. The `<h1>` tag inside `<main>` displays the product name as `<h1>Booster, 500 ml</h1>`, along with a product image. The `<div>` element with the class content acts as a container for additional content, such as the image. The `<img>` tag displays the product image with, `id="product-img"`, which identifies the image element for styling or scripting, this part specifies the location of image file `src="Images/booster_2.PNG"` and `alt="Product Image"`, provides a alternative text if the image cannot be displayed on the product page.

Table - Nutrients facts

```

61 |
62 | <p>Nutrition Facts</p>
63 | <style>
64 |   table {
65 |     border-collapse: collapse;
66 |     width: 35%;
67 |   }
68 |   th, td {
69 |     border: 1px solid black;
70 |     padding: 8px;
71 |     text-align: left;
72 |   }
73 |   th {
74 |     background-color: #f4f4f4;
75 |   }
76 | </style>
77 |
78 | <div class="table-container">
79 |   <table style="width:100%">
80 |     <tr>
81 |       <th>Nutrition</th>
82 |       <th>Amount</th>
83 |       <th>Rating</th>
84 |       <th>Recommended for adults total energy intake</th>
85 |     </tr>
86 |     <tr>
87 |       <td>Calories</td>
88 |       <td>204 kJ / 49 kcal</td>
89 |       <td><i class="fas fa-check-circle" style="color: green;"></i></td>
90 |       <td>Recommended amount is around 2,000kcal for females and 2,500kcal for males</td>
91 |     </tr>

```

Figure 5.18: Table HTML

To code this table of nutrients facts, both **HTML** and **CSS** are used to ensure a clear and visually appealing layout. **CSS** defines rules for how the table and its contents should look. The styling rules are included in `<style>`, which makes it possible to add **CSS** directly to the **HTML** document. The table is placed in a `<div class="table-container">`, which wraps the table to control its placement and styling. This `<table style="width:100%">`, helps to override the **CSS** width, so that the table fills the entire width of the container. This combination of **HTML** and **CSS** ensures, that the table is not only functional, but also visually organised and easy to read. The table is designed to adapt to different widths and highlight the headings with contrasting colours, while the cells present data with clear borders and appropriate spacing.

The vertical axis of the table is defined by header cells `<th>`, representing the categories nutrition, amount, rating, and recommendations for an adult's total energy intake. The horizontal axis is defined by rows `<tr>`, which include details like component name, amount, rating, and recommendations. As shown in figure 5.18, by using `<td>`, which represent a data cell in a table, it is possible to fill out the rows for the table. For example in the first row, the first standard cell is calories (the name of the component) and 204 kJ / 49 kcal (amount). It also includes a green icon

`<i class="fas fa-check-circle" style="color: green;"></i>` or yellow icon `<svg xmlns="http://www.w3.org/2000/svg" width="16" height="16" fill="orange"`, which informs whether the product's amount should be kept low or if it is acceptable to consume the given amount. Lastly, the recommendations section describes the suggested calorie intake for adults.

Read more - read less button

```

152 |
153 | </div>
154 | <div id="text-container">
155 |   <p>
156 |     <b>Caffeine </b> <span class="circle yellow"></span>
157 |     <span id="dots1"></span>
158 |     <span id="more1">
159 |       <br>For a typical adult the recommended amount of coffeine is 400 milligrams* a day.
160 |     </span>
161 |   </p>
162 |   <button onclick="myFunction(1)" id="myBtn1">Read more</button>
163 | </div>
164 |

```

Figure 5.19: HTML Read More

The feature depicted in figure 5.19, displays at the bottom of the product page below the table, where it is possible to access more details about the product's specific nutritional content. Starting with [HTML](#) it can be seen that the code begins with `<div id="text-container">`, which is a container that combines elements such as headings, text, and buttons for easy management and styling. Inside the container, there are several elements as follows. `<p>` contains all the text that will be displayed, however it is divided by the `<span id="more1">`, which calls the function to expand the hidden text. When using the prototype it is visualised as an 'read more' button. The read more button uses JavaScript-function `onclick="myFunction(1)"`. A JavaScript function that is called on click and passes the value 1 as a parameter. The function is used to show or hide the additional text. This will be elaborated more later regarding the 'Read More' function in JavaScript which can be found in 5.20 below. For example this section could provide more detailed information about caffeine, such as recommended amounts and a warning about overconsumption. Beyond this, `<b>` helps in emphasising the heading 'Caffeine' by making it bold. Another important element is the circle icon, which serves as a visual marker. The class "circle yellow", specifies the circle's colour is yellow, which is controlled via CSS.


```

256 // For the read more read less button
257 function myFunction(id) {
258     var dots = document.getElementById(`dots${id}`); // ($)
259     var moreText = document.getElementById(`more${id}`);
260     var btnText = document.getElementById(`myBtn${id}`);
261
262     if (dots.style.display === "none") {
263         dots.style.display = "inline";
264         btnText.innerHTML = "Read more";
265         moreText.style.display = "none";
266     } else {
267         dots.style.display = "none";
268         btnText.innerHTML = "Read less";
269         moreText.style.display = "inline";
270     }
271 }

```

Figure 5.20: JavaScript - Read more

The JavaScript function `myFunction(id)` is used to handle the 'Read more/Read less' buttons by dynamically hiding and showing additional text. The function displays or hides the extra text when a button is clicked. The `id` parameter is used to identify which section's text, button, and dots should be modified. To match the `id` parameter `getElementById`, is used to fetch the element of IDs. Backticks '``' are used to create dynamic ID names.

Therefore each button have their own ID number, based on the ID's as shown below. Using the `$` forms a string of IDs.

- **dots\$id**: → e.g., dots1, dots2, ...
- **more\$id**: → e.g., more1, more2, ...
- **myBtn\$id**: → e.g., myBtn1, myBtn2, ..

These elements represent:

- `dots`: The short version of the text are the visible dots.
- `moreText`: The hidden, longer text.
- `btnText`: The button text ("Read more" / "Read less").

```
if (dots.style.display === "none") {
```

Shows or hide the text. It checks whether dots have a display value set to "none". If dots are hidden (none), it means that the long text is already displayed.

If dots are hidden, it shows the short version and updates the buttons.

- `dots.style.display = "inline"`: Shows the short text (dots).
- `btnText.innerHTML = "Read more"`: Changes the button text to "Read more".
- `moreText.style.display = "none"`: Hides the longer text.

Otherwise, it will hide the short version and display the full text.

- `dots.style.display = "none"`: Hides the short version of the text (dots).
- `btnText.innerHTML = "Read less"`: Changes the button text to "Read less".
- `moreText.style.display = "inline"`: Displays the full text.

The function dynamically toggles between two states:

- State 1: The short version (dots) is displayed, and the button text says "Read more".
- State 2: The full text is displayed, and the button text says "Read less".

This creates a toggle effect, where the user can open and close text content by clicking the button. (W3 Schools 2024)

Circles colour code

This CSS code is used to create small circles as colour codes for the product page, which can be seen below in 5.21.

```
570  /* circles for color code on product sides */
571  .circle {
572      display: inline-block;
573      width: 16px;
574      height: 16px;
575      border-radius: 50%; /* Makes it a circle */
576      margin-left: 5px; /* Adds spacing between the text and the circle */
577      vertical-align: middle; /* Aligns with the text */
578      border: 1px solid black; /* Adds a black outline */
579  }
580
581  .circle.yellow {
582      background-color: yellow;
583  }
584
585  .circle.green {
586      background-color: green;
587  }
588
589  .circle.red {
590      background-color: red;
591  }
```

Figure 5.21: Circles colour code

The code uses `display: inline-block`, which makes the element an inline component so it can appear on the same line as text or other inline elements. At the same time, it retains a block-like structure, allowing width and height to be adjusted. Colour coding is managed through additional classes, where the background colour of the circle depends on the added class e.g., `.yellow`, `.green`, or `.red`. These classes indicate the product's status and category on the webpage.

5.5.3 My profile

This section will navigate through the 'My profile' site. Through My profile the user will be able to customise their nutrition preferences on NutriScan, based on what is important for them, when browsing groceries. The options, the user may choose from, is shown in 5.22 below.

Apply Filters

Allergies ☐ Lactose ☐ Nuts ☒ Gluten ☐ Citrus ☐ Shellfish ☐ Eggs

Nutrition label ☐ Salt ☒ Protein ☐ Carbohydrates ☐ Sugar ☐ Fat

General ☐ No needed

Apply

Clear All

• Gluten
• Protein

No filters applied.

Figure 5.22: My profile - Apply Filters

The reason for making a page where the user can choose what they want to focus on, is to make the webpage more user-based. If they have any allergies they can choose those allergies and information regarding these will pop up when choosing a grocery, which contain one of the chosen allergies. The same is the case if the user wants to focus on some specific element from the nutrition label, making it so, this element will also pop up when they are looking at different groceries. The user also have the option to choose that they do not want to focus on any filters. In the example shown above, the filters 'Gluten' and 'Protein' have been checked off in respectively the allergies and nutrition label filters list. After they have been applied, they will be shown as in the example above, and therefore be visible in the box below the 'Clear all' button. Then the user will know that their filters have been applied successfully and that their filters are active as seen in 5.23 below. If a user do not want to apply any of the nutrition label filters, or the allergies filters, then the user can apply the 'No needed' filter which then blocks off all the other filters making them inapplicable for the user until they clear all filters again.

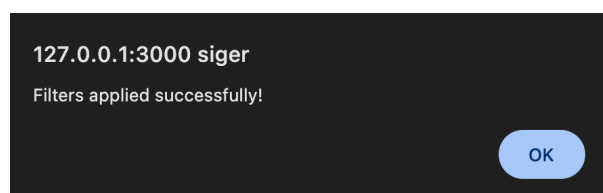


Figure 5.23: Filters applied successfully

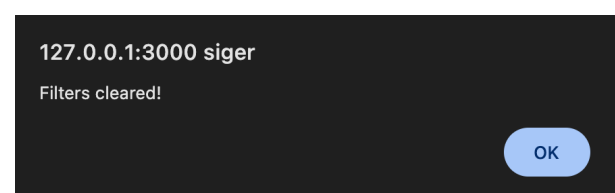


Figure 5.24: Filters cleared

When the user clicks on a filter and chooses to apply such filter, then they will get a pop-up message as seen in Figure 5.24. This message lets the user know that their choice of filters have been applied successfully. The same goes if an user chooses to clear all of their filters. Then they will get a pop-up message which is letting them know that their filters have been cleared. The reason we have chosen to build our My profile page in this way, is mainly because of the responses we got from our interviews and questionnaires. When a respondent were asked about what they where looking at on a nutrition label and how they would use a nutrition label more often, they answered as following: *"It often feels confusing—everything is written in small print, and I don't want to spend time in the store Googling unfamiliar terms. I also don't always have time to go into detail with everything on the packaging"* (appx. 5, line 424-428) and *"Maybe if it were presented in a simpler way or with colours to make it easier to understand. If I could quickly see whether something is healthy or not, it would save me time during my shopping trips"* (appx. 5, line 433-436). Another respondent said this when asked about how nutritional content could be more accessible or easier to read: *"A more visual presentation would help, like icons or colour codes indicating if something is high, low, or moderate"* (appx. 5, line 315-318). The reason we are making the 'My profile' page and the filter list is to give the user an option to easily choose if there is any aspects of their groceries, they would want to know more about. This is especially meant to support the user who don't have the time to google it in the store or if they in general think it is unmanageable to be enlightened about the nutrition label. This solution makes it easy and quick for the user to get a fast idea of what their products may contain, and hopefully further educate themselves over time.

In the following section we will delve into the [HTML](#) and JavaScript code connected to My profile.

```

40 <!-- Filter Section -->
41 <main id="main-section" class="centered-page">
42   <section id="filter-section">
43     <h2>Apply Filters</h2>
44     <form id="filter-form">
45       <h3>Allergies</h3>
46       <label><input type="checkbox" id="allergies-lactose"> Lactose</label><br>
47       <label><input type="checkbox" id="allergies-nuts"> Nuts</label><br>
48       <label><input type="checkbox" id="allergies-gluten"> Gluten</label><br>
49       <label><input type="checkbox" id="allergies-citrus"> Citrus</label><br>
50       <label><input type="checkbox" id="allergies-shellfish"> Shellfish</label><br>
51       <label><input type="checkbox" id="allergies-eggs"> Eggs</label><br>
52
53       <h3>Nutrition Label</h3>
54       <label><input type="checkbox" id="nutrition-sodium"> Sodium</label><br>
55       <label><input type="checkbox" id="nutrition-protein"> Protein</label><br>
56       <label><input type="checkbox" id="nutrition-carbohydrates"> Carbohydrates</label><br>
57       <label><input type="checkbox" id="nutrition-sugar"> Sugar</label><br>
58       <label><input type="checkbox" id="nutrition-fat"> Fat</label><br>
59
60       <h3>General</h3>
61       <label><input type="checkbox" id="general-noneeded"> No needed</label><br>
62
63       <button type="button" id="apply-filters">Apply</button>
64       <button type="button" id="clear-filters">Clear All</button>
65     </form>
66
67     <!-- Container to show the selected filters -->
68     <div id="filtered-results" style="margin-top: 20px;"></div>
69
70     <ul id="active-filters-list"></ul>
71
72   </section>
73 </main>

```

Figure 5.25: My profile, HTML

```

56 // Apply filters button logic
57 const applyButton = document.getElementById('apply-filters');
58 if (applyButton) {
59   applyButton.addEventListener('click', function(event) {
60     console.log('Apply filters button clicked!');
61
62     // Prevent default form submission behavior
63     event.preventDefault();
64
65     const resultsContainer = document.getElementById('filtered-results');
66     resultsContainer.innerHTML = ''; // Clear previous results
67
68     // Gather selected filters
69     const selectedFilters = [];
70     if (document.getElementById('allergies-lactose').checked) {
71       selectedFilters.push('Lactose');
72     }
73     if (document.getElementById('allergies-nuts').checked) {
74       selectedFilters.push('Nuts');
75     }
76     if (document.getElementById('allergies-gluten').checked) {
77       selectedFilters.push('Gluten');
78     }
79     if (document.getElementById('allergies-citrus').checked) {
80       selectedFilters.push('Citrus');
81     }
82     if (document.getElementById('allergies-shellfish').checked) {
83       selectedFilters.push('Shellfish');
84     }
85     if (document.getElementById('allergies-eggs').checked) {
86       selectedFilters.push('Eggs');
87     }

```

Figure 5.26: My profile, JavaScript 1

In the Figure 5.25, it is possible to view all of the code regarding the My profile site used in [HTML](#) to create our interface. This is where we are creating the different categories, such as seen in the top of figure 5.24. It is also where we are making the different filters to 'checkboxes', so they appear as clickable. First each filter

must be created manually. To do so, we first set the `input type="checkbox"` so the filter shows as a checkbox in the webpage, then each filter must have their own id so they can be separated from other filters. An example of this would be through `id="allergies-lactose"`. The same goes for 'Nutrition label' and 'General'.

Down below where 'General' is defined, appears two buttons, 'apply-filters' and 'clear-filters'. The applied filters will appear in the bottom of the box containing filters. In figure 5.26, as seen above. The JavaScript code is displayed, which makes the filters functional and clickable. Each of the `if` statements grants the function of making each filter box clickable and makes them look like they have been checked off as they are clicked. The snippet used in figure 5.26 is however only for 'Allergies' but the code is the same for nutrition label and general. This JavaScript snippet is also where the result container is created so that it shows and clears previous results.

```

109 // Display selected filters
110 if (selectedFilters.length > 0) {
111   const list = document.createElement('ul');
112   selectedFilters.forEach(function(filter) {
113     const listItem = document.createElement('li');
114     listItem.textContent = filter;
115     list.appendChild(listItem);
116   });
117   resultsContainer.appendChild(list);
118 } else {
119   resultsContainer.textContent = 'No filters selected.';
120 }
121 }); // Correctly close the event listener
122 }}
123 const clearButton = document.getElementById('clear-filters');
124 if (clearButton) {
125   clearButton.addEventListener('click', function() {
126     console.log('Clear all filters button clicked!');
127
128     // gather all checkboxes
129     const checkboxes = document.querySelectorAll('#filter-form input[type="checkbox"]');
130
131     // Remove the marking from all checkboxes
132     checkboxes.forEach(function(checkbox) {
133       checkbox.checked = false;
134     });
135
136     // Remove any former results
137     const resultsContainer = document.getElementById('filtered-results');
138     if (resultsContainer) {
139       resultsContainer.innerHTML = 'No filters selected.';
140     }
141   });

```

Figure 5.27: My profile, JavaScript 2

```

142 const noNeededCheckbox = document.getElementById('general-noneeded');
143 if (noNeededCheckbox) {
144   noNeededCheckbox.addEventListener('change', function() {
145     // Hent alle andre checkboxes
146     const otherCheckboxes = document.querySelectorAll('#filter-form input[type="checkbox"]:not(#general-noneeded)');
147
148     // Hvis 'No needed' er valgt
149     if (noNeededCheckbox.checked) {
150       // Gør de andre checkboxes ulikbare
151       otherCheckboxes.forEach(function(checkbox) {
152         checkbox.disabled = true;
153         checkbox.checked = false; // Fjern markering, hvis de allerede er valgt
154       });
155     } else {
156       // Gør de andre checkboxes klikbare igen
157       otherCheckboxes.forEach(function(checkbox) {
158         checkbox.disabled = false;
159       });
160     }
161   });
162 }
163

```

Figure 5.28: My profile, JavaScript 3

The two figures above, shows more JavaScript code for the My profile site and the function to choose different filters. In the top section of figure 5.27, is an extension to the previous code snippets above, which has the purpose to display the chosen filters, and at the same time if no filters are chosen then below the filter box, it will read "No filters applied". In Figure 5.28 the option to "clear all" filters is managed. In this section, a click function is added like in the other cases. In this section: `document.querySelectorAll` selects all of the checkboxes and unmarks them by setting them as `true`. Lastly the result container is cleared for all results, when clicking on the "Clear all" button. Figure 5.28 shows the "No needed" filter and how it is build. Through a lot of "if" statements, the No needed filter is build. The no needed filter has the function to disable all other filters if it is clicked upon. It will block the other filters before it is applied, and cancel out other filters if they have been added, so that No needed will be the only filter active. The `document.querySelectorAll` selects all of the filters except the No needed filter.

The other filters are disabled through `checkbox.disabled = true` and is unchecked by this `checkbox.checked = false`. If the No needed filter is unclicked again, then the other filters will be active and clickable by this line of code: `checkbox.disabled = false`.

```

166 document.addEventListener('DOMContentLoaded', function() {
167   // Apply filters button logic
168   const applyButton = document.getElementById('apply-filters');
169   if (applyButton) {
170     applyButton.addEventListener('click', function(event) {
171       event.preventDefault(); // Prevent form submission
172       console.log('Apply filters button clicked!');
173
174       const selectedFilters = [];
175
176       // Gather selected filters from checkboxes
177       const checkboxes = document.querySelectorAll('#filter-form input[type="checkbox"]');
178       checkboxes.forEach((checkbox) => {
179         if (checkbox.checked) {
180           selectedFilters.push(checkbox.id);
181         }
182       });
183
184       // Save selected filters in localStorage
185       localStorage.setItem('selectedFilters', JSON.stringify(selectedFilters));
186       console.log('Filters saved:', selectedFilters);
187
188       // Optionally, show a confirmation to the user
189       alert('Filters applied successfully!');
190     });
191   }

```

Figure 5.29: My profile, JavaScript 4

```

193 // Apply filters when the page loads
194 const storedFilters = JSON.parse(localStorage.getItem('selectedFilters')) || [];
195 const activeFiltersList = document.getElementById('active-filters-list');
196 if (storedFilters.length > 0) {
197   storedFilters.forEach((filter) => {
198     const listItem = document.createElement('li');
199     listItem.textContent = filter.replace(/-/g, ' '); // Remove hyphens for readability
200     activeFiltersList.appendChild(listItem);
201   });
202 } else if (activeFiltersList) {
203   activeFiltersList.textContent = 'No filters applied.';
204 }

```

Figure 5.30: My profile, JavaScript 5

In figure 5.29 is the code for making the webpage able to save and store the selected filters. Said filter remain as saved and active, regardless if the webpage is refreshed. The `selectedFilters` array is through `JSON.stringify` converted to a JSON string for storage in the browsers local storage. A `console.log` displays the saved filters for debugging. At the bottom of Figure 5.29 is the alert snippet which pops up when the user have applied their filters, with a message which replies "Filters applied successfully!". The top part of figure 5.30 is responsible for applying the saved and stored filters when the page loads. When the page loads, it will check the `localStorage` for previously saved filters. If there is already saved filters, they will be visibly shown in the active filter list. If no filters have been applied or saved, it will read "No filters active".

6 Reflection and discussion

6.1 Evaluation of our web solution

In this chapter, we seek to evaluate the data collection methods with focus on, how this has affected our process. Through this, we seek to evaluate the process of finding requirements within the data, as well as the prioritisation of these. Secondly, we will evaluate to what degree the requirements has been fulfilled, as well as reflect upon how we can further work to implement said unfulfilled requirements.

6.1.1 Reflection on methods

Through this project, we have utilised both interviews, as well as a questionnaire. While the data collection were vaster than first anticipated, we argue that these collection methods have been able to complement one another well. Our preliminary questionnaire led us to the ability to ask questions, we had not yet thought of ourselves. We are however aware of the fact, that the questionnaire was primarily sent to students of the Department of Computer Science. This focus may have introduced a degree of selection bias, thereby constraining the representativeness of our findings and falling short of the holistic perspective we initially aimed to achieve. Furthermore, some responses appeared to reinforce pre-existing assumptions, potentially amplifying confirmation bias and influencing our initial interpretations of system requirements. By engaging with participants in open-ended discussion, we were able to gain nuanced insights into the needs of potential users of our solution. The interviews therefore served as a crucial counterbalance to our data collection, enabling us to more critically evaluate the patterns we found within the answers of our questionnaire, and in a combination of both, securing a more comprehensive foundation for the development of our system requirements.

6.1.2 Fulfilment of requirements

Based on the data we have collected, and described in chapter 3.1 and 3.2, we have made 12 system requirements, which we prioritised according to the MoSCoW method. In this chapter we seek to evaluate the overall fulfilment of said requirements, as well as any shortcomings along with suggestions as to how they could have been fulfilled to a further degree. To classify a requirement as being either completely or partly fulfilled, we argue that it must be able to demonstrate the functionalities intended. This will be specifically explained for each requirement through the specification column of figure 6.1. The figure overall shows an overview of how we have evaluated the fulfilment of NutriScan's system requirements.

MoSCoW priority	No.	Function	Fulfilment	Specification
Must have	R1	Overview of macro nutrition	Fulfilled	Data displayed when product is searched
	R2	Overview of micro nutrition	Fulfilled	Data displayed when products is searched
	R6	Users' ability to view in depth information	Fulfilled	Data displayed through table, with ability to 'Read More' with onclick button.
	R9	Showcasing nutrition and ingredient information	Fulfilled	Displayed in table, alongside indication of rating
	R10	Differentiate known problematic ingredients	Somewhat fulfilled	Examples provided in displayed data, further implementation requires more advanced dataset and database containing custom lists
Should have	R3	Easy to understand data	Somewhat fulfilled	Data we see relevant is displayed, but requires further user feedback to confirm levels of understanding
	R4	Product comparison	Not fulfilled	
	R7	Scan/search function	Somewhat fulfilled	Search function fully implemented through JavaScript – missing search function, which was deemed out of our scope currently
	R11	User defined prioritisation	Somewhat fulfilled	User prioritisation achieved through 'My Profile', yet is missing the final link between local storage of user and product.
	R12	Identifying ultra processed foods	Not fulfilled	Could be solved with in depth data set and database, from which ingredients could be classified within a spectrum of being processed.
Could have	R5	Interesting visualisation of data	Somewhat fulfilled	The rating system is implemented within the data table, but not fully linked between user prioritisation as we had hoped to do.
	R8	Comparisons based on product price	Not fulfilled	

Figure 6.1: Fulfilment of system requirements

The Must have requirements are of course deemed to be of the highest level of importance, hence we will delve into these first. Upon further examination, we see that requirements *R1*, *R2* and *R9* all had similar implementation descriptions in figure 3.6. The display of information, such as specifically micro- and macronutrition, as well as the ingredient lists, are of the utmost priority for our system to fulfil its purpose. We argue, that this need is being met, as the user is presented with all needed information to make informed choices about their consumption, yet we also acknowledge that the lack of a proper database complicates the task of further expand the system. We also see, that given the similarities in the solution for the requirements, that we could have been more specific when defining what in fact would constitute as a different requirement. Besides this, we have *R10*, a Must have requirement, that we deem to only somewhat have been able to fulfil. We argue that we have fulfilled the core idea of what should be demonstrated, with

the description of problematic ingredients, however, this suffers more greatly than the average data from the lack of a database. This is based upon the Should have requirement *R11*, where the user should be able to define their own preferences. This is functional, but we have yet to find the way to link the local storage from the user to the product, an issue we imagine could have been solved with a functioning database.

Some aspects we have actively decided to postpone, is the tool of comparison, both between health benefits, as well as the comparison in economic aspects. This is the reason that both *R4* and *R8* are considered not fulfilled, as this was deemed too far off the core concept of NutriScan, and therefore set to be worked upon in further work.

6.1.3 Future work

While the current implementation of NutriScan fulfils several key requirements, there remain areas for improvement and future development to enhance the system's functionality and usability. Below, we outline proposed future work to address the not fulfilled requirements and further refine the solution.

One of the main planned additions is integrating a barcode scanning feature into the web solution. This enhancement would allow users to scan product barcodes directly, making the process of retrieving nutritional information faster and easier. This feature addresses the not fulfilled requirements related to improving accessibility and usability in *R7*. Automating the input process will presumably significantly improve the user experience, especially for those who find typing product names inconvenient or are prone to errors. The barcode scanning functionality could be implemented using a camera-based API or a third-party library to ensure compatibility across devices.

Another key area for future work, involves usability testing. After a prototype of the solution is finalised, potential users could be invited to interact with the system to identify potential flaws, validate the design, and assess its usability and functionality. External participants, who are not as familiar with the system, can provide impartial feedback, highlighting areas of confusion or inefficiency that may not be apparent to us as the development team (Garrett 2011). This iterative testing approach will presumably ensure that NutriScan becomes more intuitive and user-centred over time. After analysing the feedback, a revised prototype would be developed and subjected to more rounds of testing to confirm improvements and address any remaining issues.

Additionally, requirements *R4* and *R8*, which focus on tools for comparing products based on health benefits and economic aspects, would be a rather large part of our implementation for future additions. These features will give the users the ability to make more informed decisions by making comparisons of product nutritional profiles and prices available, both within a singular and differing stores. These additional functionalities will require an integration of an expanded database and a comparison of algorithms to make sure that the information presented is both comprehensive and as well as accurate. In the following figure 6.2 is a code snippet from JavaScript, which is an extension of the code used in My profile.

```
206 // Apply filters to products
207 const products = document.querySelectorAll('.product-item');
208 products.forEach((product) => {
209   const productTags = product.dataset.tags ? product.dataset.tags.split(',') : [];
210   const isVisible = storedFilters.every((filter) => productTags.includes(filter));
211
212   if (isVisible || storedFilters.length === 0) {
213     product.style.display = 'block'; // Show product if it matches filters
214     const matchesFilters = storedFilters.some(filter => productTags.includes(filter));
215     if (matchesFilters) {
216       product.classList.add('filtered'); // Highlight product if it contains a selected filter
217     } else {
218       product.classList.remove('filtered'); // Remove highlight if no match
219     }
220   } else {
221     product.style.display = 'none'; // Hide product if it doesn't match any filters
222   }
223 });
```

Figure 6.2: My profile - JavaScript 6

In figure 6.2 is the JavaScript code, which should work as the link between My profile and the product list. With this code, it should be possible to apply the different filters, that one wish to apply, and then those filters should be active and visible for the user, when entering the product list. If a user applied protein in nutrition label and gluten in allergies as filters, then when entering a product, an alert or another visible notification, should notify the user that the product they are looking at, contains protein and gluten. The code is consists of product tags. Each product would get its own tag, and then manually we would write in those product tags, with the different ingredients of the product. Then when a user would click on a product, the product tag would be connected to the filters and be shown to the user. As for now, the code shown in Figure 6.2 does not work, which is why it is placed in future works. Our intention with this, is to be able to connect these dots and fix the issue in the future.

By implementing these aspects, NutriScan can evolve into a more integrated and user-friendly tool, fulfilling its purpose of helping users make informed decisions about their consumption and dietary habits.

7 Conclusion

We have throughout this project worked with the following problem statement:

How can a web-based solution assist consumers in understanding and navigating within the scientific language of nutritional labels, while offering personal preferences and restrictions that align with dietary needs?

The existing research, used within this project, indicated a shift from diets based on unprocessed foods to (ultra)processed foods. This can be alarming as it increases the risks of malnutrition, overweight or underweight, and cardiovascular diseases. In an attempt to rectify this, nutrition labels have been mandated on products since around 2016. However, the existing research indicated that a larger amount of people tend to find nutrition labels confusing, difficult to understand, and time consuming. Some of these tendencies were also found in our data, but these also conveyed an interest in nutrients and ingredients, ecology, e-numbers, and artificial products among others. These insights motivated our interest in the problem area.

In order to answer our problem statement, we have gathered information from possible users in the age group 15-30 years old, both through a questionnaire as well as interviews, which has helped us map out different requirements. Through the concept of Computational Thinking we narrowed the problem area to be able to get a more specific vision of our web solution and how we could meet the requirements. These requirements have helped us build prototypes which formed our web solution, NutriScan, in the hopes of accommodating the user's different needs, helping them select enlightened choices concerning groceries.

To assist consumers in understanding and navigating the labels, we aimed to make the web solution's interface simple and manageable. The purpose is to make it easy to either use the search or picture carousel functionalities to get to a specific product quickly and get an overall idea of the content. This is to accommodate the possibility of using the web solution while at the grocery store, but still have the ability to view more information if wanted as well as at a different time. To offer more personalised information and hopefully make the web solution more attractive and relevant to use, we created the profile page with the filter functionality including allergies and nutrition. Through these tools, the user will not only be able to view information on products, but also have personal dietary needs highlighted according to the users preferences.

With this in mind, we find it relevant to reflect upon the challenges we have experienced, such as our difficulties with finding a balance between simplicity and usability. In addition to this, we also found it challenging to strike a balance between providing enough information for the user to actually be able to understand and use the information, more than already given on the products, but still keep it simple. Furthermore, we acknowledge the challenges of attempting to meet all requirements we put forth, but find it valuable to reflect upon how somewhat fulfilled requirements still add value to our process and product.

The findings and reflections presented in this report provide a base for the further development of the solution. Moving forward, usability testing, iterative design, and the integration of features, such as barcode scanning for real-time product information, would be beneficial as the next steps, to improve the web solution's accessibility, usability, and effectiveness.

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